United States Patent	12416030
Kind Code	B2
Date of Patent	September 16, 2025
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Identification of an α-1,2-fucosyltransferase for the in vivo production of pure LNFP-I

Abstract

The present disclosure discloses the identification and introduction of a specific heterologous gene (denoted as smob), which encodes an α-1,2-fucosyltransferase, into an LNT production strain to produce LNFP-I in particular. The smob gene originates from the organism *Sulfuriflexus mobilis* (https://www.dsmz.de/collection/catalogue/details/culture/DSM-102939), which is a sulfur-oxidizing bacterium isolated from a brackish lake sediment.

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Appl. No.:	**18/561170**
Filed (or PCT Filed):	**May 17, 2022**
PCT No.:	**PCT/EP2022/063315**
PCT Pub. No.:	**WO2022/243312**
PCT Pub. Date:	November 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240417766 A1	Dec. 19, 2024

Foreign Application Priority Data

DK	PA202170250	May. 17, 2021

Publication Classification

Int. Cl.: **C12P19/04** (20060101); **C12N1/20** (20060101); **C12N9/10** (20060101); **C12N9/26** (20060101); **C12N15/70** (20060101); **C12P19/18** (20060101); **C12R1/19** (20060101)

U.S. Cl.:

CPC **C12P19/04** (20130101); **C12N1/20** (20130101); **C12N9/1051** (20130101); **C12N9/2431** (20130101); **C12N15/70** (20130101); **C12P19/18** (20130101); C12R2001/19 (20210501); C12Y204/01122 (20130101); C12Y204/01146 (20130101); C12Y302/01026 (20130101)

Field of Classification Search

CPC: C12P (19/04); C12P (19/18); C12N (1/20); C12N (9/1051); C12N (9/2431); C12N (15/70); C12N (9/1048); C12N (15/09); C12R (2001/19); C12Y (204/01122); C12Y (204/01146); C12Y (302/01026); C12Y (204/01069); C07H (3/06)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a National Stage entry pursuant to 35 U.S.C. § 371 of International Application No. PCT/EP2022/063315, filed on May 17, 2022, which claims priority to Denmark Application No. PA202170250, filed on May 17, 2021, the entire contents of all of which are hereby incorporated by reference in their entirety.
SEQUENCE LISTING
(2) This instant application contains a sequence listing which has been submitted in a ascii text file

via Patent Center and is hereby incorporated by reference in its entirety. Said text file, created on Nov. 14, 2023, is named 032991-8008 sequence listing.txt, and is 68,453 bytes in size.

FIELD

(3) The present disclosure discloses the identification and introduction of a specific heterologous gene, which encodes an α-1,2-fucosyltransferase, into an LNT production strain to produce LNFP-I as the major HMO compound.

BACKGROUND

(4) To bypass the drawbacks associated with the chemical synthesis of HMOs, several enzymatic methods and fermentative approaches have been developed. Fermentation based processes have been developed for several HMOs such as 2′-fucosyllactose, 3-fucosyllactose, lacto-N-tetraose, lacto-N-neotetraose, Lacto-N-fucopentaose I, 3′-sialyllactose and 6′-sialyllactose. Fermentation based processes typically utilize genetically engineered bacterial strains, such as recombinant *Escherichia coli* (*E. coli*).

(5) Biosynthetic production, such as a fermentation process, of HMOs is a valuable, cost-effective and large-scale applicable solution for HMO manufacturing. It relies on genetically engineered bacteria constructed so as to express the glycosyltransferases needed for synthesis of the desired oligosaccharides and takes advantage of the bacteria's innate pool of nucleotide sugars as HMO precursors.

(6) Recent developments in biotechnological production of HMOs have made it possible to overcome certain inherent limitations of bacterial expression systems. For example, HMO-producing bacterial cells may be genetically engineered to increase the limited intracellular pool of nucleotide sugars in the bacteria (WO2012112777), to improve activity of enzymes involved in the HMO production (WO2016040531), or to facilitate the secretion of synthesized HMOs into the extracellular media (WO2010142305, WO2017042382).

(7) Further, expression of genes of interest in recombinant cells may be regulated by using particular promoter sequences or other gene expression regulators, like e.g. what has recently been described in WO2019123324.

(8) Recently, improvements were made in the production of fucosylated HMOs, where a number of fucosyltransferases were identified, which showed enhanced functionality, thus enhancing the amount of produced product, as described in WO2019008133. In example, *E. coli* expressing a selection of α-1,2-fucosyltransferases, was shown in WO2019008133 to be capable of producing LNFP-I, however with considerable side product HMOs.

(9) Thus, there is still an unmet need for alternative genetic engineering to produce HMOs.

SUMMARY

(10) Herein we disclose an α-1,2-fucosyltransferase, Smob, which is superior to the previously disclosed α-1,2-fucosyltarsnferases for the biosynthetic production of LNFP-I, which overcomes the technical issue of side product formation and increases the LNFP-I titer and yield.

(11) The expression of Smob, may in turn be combined with further genetic modifications, such as a sucrose utilization system or one or more metabolic pathway modifications that facilitate product formation and increase the titer of LNFP-I, while maintaining a low level of byproducts.

(12) The present disclosure firstly relates to a method for producing one or more fucosylated human milk oligosaccharide (HMO), the method comprising the steps of, a) providing a genetically engineered cell capable of producing an HMO, wherein said cell comprises a recombinant nucleic acid encoding an α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 70% identical to SEQ ID NO: 1, b) culturing the cell according to (a) in a suitable cell culture medium to express said nucleic acid; and c) harvesting the HMO(s) produced in step (b).

(13) The disclosure further relates to a genetically engineered cell comprising a recombinant nucleic acid sequence encoding an α-1,2-fucosyltransferase protein as shown in SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 70% identical,

such as at least 75% identical, or such as at least 80% identical to SEQ ID NO: 1.
(14) The disclosure also relates to a nucleic acid construct comprising a recombinant nucleic acid sequence which is at least 70% identical to SEQ ID NO: 2.
(15) The disclosure further relates to the use of a genetically engineered cell, or the nucleic acid construct as described herein, in the manufacturing of one or more HMOs.

DETAILED DESCRIPTION

(16) The steps of any method disclosed herein do not have to be performed in the exact order disclosed, unless explicitly stated.
(17) The present disclosure relates to method for producing fucosylated oligosaccharides, wherein a genetically engineered cell is used for producing said fucosylated oligosaccharide. Said genetically engineered cell has been genetically engineered to express a heterologous α-1,2-fucosyltransferase which is capable of transferring a fucose residue from a donor substrate to an acceptor molecule, wherein said α-1,2-fucosyltransferase has higher specificity for lacto-N-tetraose (LNT) as acceptor molecule than towards lactose as acceptor molecule.
(18) The α-1,2-Fucosyltransferase—Smob
(19) The present disclosure refers to the identification of a specific heterologous gene (denoted as smob), which codes for an α-1,2-fucosyl-transferase, and its introduction into an LNT production strain in order to produce LNFP-I. The smob gene originates from the organism *Sulfuriflexus mobilis* (https://www.dsmz.de/collection/catalogue/details/culture/DSM-102939), which is a sulfur-oxidizing bacterium isolated from a brackish lake sediment.
(20) In a preferred exemplary embodiment, the amino acid sequence encoding the α-1,2-fucosyltransferase Smob is 100% identical to SEQ ID NO: 1, which is the GenBank ID WP_126455392.1 originating from *Sulfuriflexus mobilis*.
(21) Contrary to other α-1,2-fucosyltransferases, such as FutC (Genbank ref. no: WP_080473865.1), the Smob α-1,2-fucosyltransferase is able to selectively fucosylate LNT, while it has a very low activity on lactose. These beneficial effects have been confirmed in deep-well assays, see FIG. **1-5**, as well as in fermentation data in the Examples below. Provided herein are α-1,2-fucosyltransferases originating from bacterial cells. Said fucosyltransferases utilizes lacto-N-tetraose as the acceptor molecule for their fucosyltransferase activity. Said fucosyltransferases can be used to synthesize fucosylated oligosaccharides based on LNT as acceptor molecule.
(22) In this manner, not only higher LNFP-I titers can be achieved with Smob compared to FutC or FucT54 (FIG. **1**A), but the pronounced formation of 2′-FL, which is a result of low specificity of the FutC enzyme, can be avoided (FIG. **2**).
(23) As shown in FIG. **3-5**, the obtained final HMO profile differs significantly when the FutC, FucT54 or Smob enzymes are introduced in an LNT-producing strain. Notably, the HMO profile obtained from smob-expressing cells consists almost exclusively of LNFP-I (91%) (FIG. **5**).
(24) The features described here for the Smob enzyme are very interesting in the sense that a microorganism such as *Sulfuriflexus mobilis*, which is not a human pathogen, possesses an enzyme that can have such an activity on HMOs. Even more interesting, the expression of the Smob enzyme in *E. coli* cells can result in much higher LNFP-I titers compared to other α-1,2-fucosyltransferases, such as the FucT54 enzyme (Genbank ref. no: WP_013031010.1, also disclosed in WO2019/008133), when expressed in an identical strain background and under the control of the same regulatory elements and at the same gene dosage.
(25) Also, one could expect that the ability to specifically fucosylate LNT and not lactose is a result of an enzyme (e.g. FutC) engineering effort. However, a wild-type enzyme produced by a naturally occurring microorganism that is found in an otherwise unexpected habitat, such as a brackish lake sediment, can show a desired fucosylation specificity towards an HMO molecule, i.e., LNT.
(26) An α-1,2-fucosyl-transferase is responsible for adding a fucose onto the galactose residue of the O-antigen repeating unit via an α-1,2 linkage. The sequence of the protein may be altered without losing the functionality.

(27) The functionality of the Smob α-1,2-fucosyl-transferase is the transfer of a fucosyl moiety from a donor molecule onto a Gal moiety of an acceptor through an α-1,2 coupling. The donor is e.g. GDP-Fucose, if the acceptor molecule is lactose, LNT and/or LNFP-I.
(28) Thus, in one or more exemplary embodiments, the Smob α-1,2-fucosyltransferase or a functional homologue thereof transfers a fucosyl group from GDP-Fucose onto lactose and/or LNT, thus generating 2′-FL and/or LNFP-I (see FIG. **10**). In a similar manner, the enzyme transfers a fucosyl group from GDP-Fucose onto lactose, thus generating 2′-FL.
(29) The activity of the Smob α-1,2-fucosyltransferase is higher for the fucosylation of LNT than for lactose.
(30) The Smob α-1,2-fucosyltransferase has a low activity on lactose and a high activity on LNT. Examples of high activity of the Smob α-1,2-fucosyltransferase on LNT and low activity of said α-1,2-fucosyltransferase on lactose is shown in Example 1.
(31) In one or more exemplary embodiments, the recombinant nucleic acid encodes an α-1,2-fucosyltransferase protein as shown in SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 70% identical to SEQ ID NO: 1, such as at least 71% identical, at least 72% identical, at least 73% identical, at least 74% identical, at least 75% identical, at least 76% identical, at least 77% identical, at least 78% identical, at least 79% identical, at least 80% identical, at least 81% identical, at least 82% identical, at least 83% identical, at least 84% identical, at least 85% identical, at least 86% identical, at least 87% identical, at least 88% identical, at least 89% identical, at least 90% identical, at least 91% identical, at least 92% identical, at least 93% identical, at least 94% identical, at least 95% identical at least 96% identical, at least 97% identical, at least 98% identical, or at least 99% identical.
(32) In one or more presently preferred exemplary embodiments, the amino acid sequence is at least 75% identical to SEQ ID NO: 1.
(33) Sequence Identity
(34) The term "sequence identity of [a certain] %" in the context of two or more nucleic acid or amino acid sequences means that the two or more sequences have nucleic acids or amino acid residues in common in the given percent, when compared and aligned for maximum correspondence over a comparison window or designated sequences of nucleic acids or amino acids (i.e. the sequences have at least 90 percent (%) identity). Percent identity of nucleic acid or amino acid sequences can be measured using a BLAST 2.0 sequence comparison algorithm with default parameters, or by manual alignment and visual inspection (see e.g. http://www.ncbi.nlm.nih.gov/BLAST/). This definition also applies to the complement of a test sequence and to sequences that have deletions and/or additions, as well as those that have substitutions. An example of an algorithm that is suitable for determining percent identity, sequence similarity and for alignment is the BLAST 2.2.20+ algorithm, which is described in Altschul et al. *Nucl. Acids Res.* 25, 3389 (1997). BLAST 2.2.20+ is used to determine percent sequence identity for the nucleic acids and proteins of the disclosure. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (http://www.ncbi.nlm.nih.gov/). Examples of commonly used sequence alignment algorithms are CLUSTAL Omega (www.ebi.ac.uk/Tools/msa/clustalo/), EMBOSS Needle (www.ebi.ac.uk/Tools/psa/emboss_needle/), MAFFT (mafft.cbrc.jp/alignment/server/), or MUSCLE (www.ebi.ac.uk/Tools/msa/muscle/).
(35) Preferably, the sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, J. Mol. Biol. 48:443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, Trends Genet. 16:276-277), preferably version 5.0.0 or later (available at https://www.ebi.ac.uk/Tools/psa/emboss needle/). The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of 30 BLOSUM62) substitution matrix. The output of Needle labeled "longest identity" (obtained

using the -nobrief option) is used as the percent identity and is calculated as follows: (Identical Residues×100)/(Length of Alignment−Total Number of Gaps in Alignment).

(36) Preferably, the sequence identity between two nucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1 970, supra) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, Trends Genet. 16:276-277), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the DNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows: (Identical Deoxyribonucleotides×100)/(Length of Alignment−Total Number of Gaps in Alignment).

(37) Functional Homologue

(38) A functional homologue of a protein/nucleotide as described herein is a protein/nucleotide with alterations in the genetic code, which retain its original functionality. A functional homologue may be obtained by mutagenesis. The functional homologue should have a remaining functionality of at least 50%, such as 60%, 70%, 80%, 90% or 100% compared to the functionality of the protein/nucleotide.

(39) A functional homologue of any one of the disclosed amino acid sequences can also have a higher functionality. A functional homologue as described herein is able to participate in the HMO production, in terms of HMO yield, purity, reduction in biomass formation, viability of the genetically engineered cell, robustness of the genetically engineered cell, or reduction in consumables.

(40) A Method for Producing One or More Human Milk Oligosaccharide (HMO)

(41) When applying the Smob α-1,2-fucosyltransferases in industrial production, one is able to achieve higher conversion rates in the enzymatic production of complex HMOs.

(42) Thus, in one or more exemplary embodiments, the present disclosure relates to a method for producing one or more human milk oligosaccharide (HMO) comprising the steps of, a) providing a genetically engineered cell capable of producing an HMO, wherein said cell comprises a recombinant nucleic acid encoding an α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 70% identical to SEQ ID NO: 1, b) culturing the cell according to (a) in a suitable cell culture medium to express said nucleic acid; and c) harvesting the HMO(s) produced in step (b).

(43) In one embodiment, the one or more HMOs contains a least one fucosylated HMO, preferably selected from the group consisting of 2′-FL, LNFP-I and DFL.

(44) In another embodiment the one or more HMOs contains a least one fucosylated HMO, preferably selected from the group consisting of 2′-FL, LNFP-I and LNDFH-I.

(45) In a preferred embodiment the method predominantly produces LNFP-1, meaning that more than 70% of the total HMO produced is LNFP-I. In the context of the present invention LNFP-I can also be termed the HMO product or major HMO.

(46) Human Milk Oligosaccharide (HMO)

(47) In the context of the disclosure, the term “oligosaccharide” means a saccharide polymer containing a number of monosaccharide units. In some embodiments, preferred oligosaccharides are saccharide polymers consisting of three, four, five or six monosaccharide units, i.e. trisaccharides, tetrasaccharides, pentasaccharides or hexasaccharides. Preferable oligosaccharides of the disclosure are human milk oligosaccharides (HMOs).

(48) The term “human milk oligosaccharide” or “HMO” in the present context means a complex carbohydrate found in human breast milk. The HMOs have a core structure comprising a lactose unit at the reducing end that can be elongated by one or more β-N-acetyl-lactosaminyl and/or one or more β-lacto-N-biosyl units, and this core structure can be decorated by one or more α-L-fucopyranosyl and/or α-N-acetyl-neuraminyl (sialyl) moieties.

(49) In this regard, the non-acidic (or neutral) HMOs are devoid of a sialyl residue, and the acidic HMOs have at least one sialyl residue in their structure. The non-acidic (or neutral) HMOs can be fucosylated or non-fucosylated. Examples of such neutral non-fucosylated HMOs include lacto-N-triose II (LNT-II) lacto-N-tetraose (LNT), lacto-N-neotetraose (LNnT), lacto-N-neohexaose (LNnH), para-lacto-N-neohexaose (pLNnH), para-lacto-N-hexaose (pLNH) and lacto-N-hexaose (LNH). Examples of neutral fucosylated HMOs include 2′-fucosyllactose (2′-FL), lacto-N-fucopentaose I (LNFP-I), lacto-N-difucohexaose I (LNDFH-I), 3-fucosyllactose (3′-FL), difucosyllactose (DFL), lacto-N-fucopentaose II (LNFP-II), lacto-N-fucopentaose III (LNFP-III), lacto-N-difucohexaose III (LNDFH-III), fucosyl-lacto-N-hexaose II (FLNH-II), lacto-N-fucopentaose V (LNFP-V), lacto-N-difucohexaose II (LNDFH-II), fucosyl-lacto-N-hexaose I (FLNH-I), fucosyl-para-lacto-N-hexaose I (FpLNH-I), fucosyl-para-lacto-N-neohexaose II (F-pLNnH II) and fucosyl-lacto-N-neohexaose (FLNnH). Examples of acidic HMOs include 3′-sialyllactose (3′-SL), 6′-sialyllactose (6′-SL), 3-fucosyl-3′-sialyllactose (FSL), 3′-O-sialyllacto-N-tetraose a (LST a), fucosyl-LST a (FLST a), 6′-O-sialyllacto-N-tetraose b (LST b), fucosyl-LST b (FLST b), 6′-O-sialyllacto-N-neotetraose (LST c), fucosyl-LST c (FLST c), sialyl-lacto-N-hexaose (SLNH), sialyl-lacto-N-neohexaose I (SLNH-I), sialyl-lacto-N-neohexaose II (SLNH-II) and disialyl-lacto-N-tetraose (DS-LNT).
(50) In the context of the present disclosure lactose is not regarded as an HMO species.
(51) One or More HMO(s)
(52) By the term “one or more HMOs” is meant that an HMO production cell may be able to produce a single HMO structure (a first HMO) or multiple HMO structures (a second, a third, etc. HMO). Multiple HMO structures from one cell is in particular the case when the cell contains more than one glycosyl transferase, since this often leads to HMO by-product formation. Typically, by-product HMOs are either the major HMO precursors or products of further modification of the major HMO. In some embodiments, it may be desired to produce the product HMO in predominant amounts and by-product HMOs in minor amounts. Cells and methods for HMO production described herein allow for controlled production of an HMO product with a defined HMO profile, e.g., in one embodiment, the produced HMO mixture wherein the product HMO is a dominating HMO compared to the other HMOs (i.e. by-product HMOs) of the mixture, i.e. the product HMO is produced in higher amounts than other by-product HMOs; in other embodiments, the cell producing the same HMO mixture may be tuned to produce one or more by-product HMOs in higher amount than product HMO. For example, during the production of the major HMO LNFP-I, often a significant amount of 2′-FL and some DFL, LNT-II, LNT, and potentially also LNnH, parα-LNH are present after fermentation, these can be considered by-product HMO, but may also be desired as part of the final HMO product. With the genetically modified cells of the present invention the level of 2′-FL in the LNFP-I product can be significantly reduced.
(53) Human Milk Oligosaccharide (HMO) Blend The term “blend” or “HMO blend” refers to a mixture of two or more HMOs and/or HMO precursors, such as but not limited to HMOs selected from LNT-II, LNT, LNnT, LNH, LNnH, p-LNH, p-LNnH, 2′-FL, 3FL, DFL, LNFP-I, LNFP-II, LNFP-III, LNFP-V, F-LNnH, DF-LNH I, DF-LNH II, DF-LNH I, DF-para-LNH, DF-para-LNnH, 3′-SL, 6′-SL, FSL, F-LST a, F-LST b, F-LST c, LST a, LST b, LST c and DS-LNT.
(54) In some exemplary embodiments, it may be preferred that a genetically engineered cell produces a single HMO, in other preferred exemplary embodiments, a genetically engineered cell producing multiple HMO structures may be preferred.
(55) In some exemplified embodiments, it may be desired to produce the predominant HMO or HMOs in abundant amounts and by-product HMOs in minor amounts.
(56) In one or more exemplary embodiments, the method produces one or more fucosylated human milk oligosaccharide(s) (HMO(s)). In the context of the present invention this means comprising one or more fucosylated HMOs and it is understood that the production of one or more fucosylated HMOs does not rule out the presence of other HMO's such as neutral core HMO's, like LNT-II and

LNT produced as by-product HMOs in the fermentation process.
(57) In one or more exemplary embodiments, the method may produce one or more HMOs selected from the group consisting of 2′-FL, LNT-II, LNT, LNFP-I, LNDFH-I and DFL.
(58) In one or more exemplary embodiments, the method may produce one or more fucosylated HMOs selected from the group consisting of 2′-FL, LNFP-I, LNDFH-I and DFL. Preferably, at least LNFP-I is present.
(59) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the one or more HMOs are 2′-FL, DFL, LNT-II, LNT and LNFP-I.
(60) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the one or more HMOs are 2′-FL, DFL, LNT and LNFP-I.
(61) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the one or more HMOs are LNFP-I and potentially one or more of 2′-FL, LNT-II and/or LNT.
(62) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the one or more HMOs are LNFP-I and potentially one or more of 2′-FL, DFL and/or LNT.
(63) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the one or more HMOs are LNFP-I and LNT.
(64) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the one or more HMOs are LNFP-I and 2′-FL.
(65) Predominant HMO
(66) The term “predominant” is used herein to define a single HMO species being more than 70 molar % of the total amount of harvested HMOs, such as but not limited to more than 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.5%. The same definition applies to a blend of HMOs, meaning that a blend of for example two HMOs are “predominant”, when the blend is more than 70% of the total amount of harvested HMOs.
(67) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the predominant HMOs produced are LNFP-I, 2′-FL and/or LNT.
(68) In one or more exemplary embodiments, the method produces one or more HMOs, wherein the predominant HMO produced is LNFP-I. Specifically, LNFP-I is produced in more than 70 molar % of the total HMO, such as more than 75%, such as more than 80%, such as more than 85%, such as more than 90%, such as more than 95 molar % of the total HMO.
(69) In one or more exemplary embodiments, the method produces one or more HMOs, wherein the predominant HMO produced is a blend of LNT and LNFP-I.
(70) In one or more exemplary embodiments, the method may produce one or more HMO, wherein the predominant HMOs produced is a blend of 2′-FL and LNFP-I.
(71) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the predominant HMOs produced is a blend of LNFP-I and LNT-II.
(72) LNFP-I as the Predominant HMO
(73) One objective with the present disclosure is to achieve higher LNFP-I titers, while GDP-fucose and lactose are not exclusively used in the 2′-FL biosynthetic route, but remain available for and are directed to the LNFP-I route instead. This facilitates the development of an industrial process towards the production of LNFP-I by fermentation, since the HMO profile acquired in the end of the fermentation consists almost exclusively of LNFP-I. One advantage of the method described herein is therefore that production of LNFP-I by fermentation is enabled with only minimal levels of 2-′FL being formed by the producing cell.
(74) In one or more presently preferred exemplary embodiments, the predominant HMO produced is LNFP-I. Specifically, LNFP-I is produced in more than 70 molar % of the total HMO, such as more than 75%, such as more than 80%, such as more than 85%, such as more than 90%, such as

more than 95 molar % of the total HMO.
(75) Lacto-N-fuco-pentaose I (LNFP-I) is a pentasaccharide, more precisely, a neutral fucosylated pentasaccharide composed of fucose, galactose, N-acetylglucosamine, galactose, and glucose (Fucα1-2Galβ1-3GlcNAcβ1-3Galβ1-4Glc). It is naturally present in human milk. LNFP-I supports immune health via inhibition of pathogen adherence to the intestinal cell wall and antimicrobial effects via binding to toxins. LNFP-I also has a positive impact on growth of bifidobacteria, which are beneficial for gut health.
(76) The production of predominantly LNFP-I is particularly challenging in large scale, yet the present disclosure provides an effective means for the production of predominantly LNFP-I.
(77) As seen from the Examples section and in particular FIG. **5**, the Smob α-1,2-fucosyltransferase shows high specificity for LNT and simultaneously very low specificity for lactose, as revealed by the high-level LNFP-I production and the very low titers of other HMOs detected in smob-expressing cells. Thus, the genetically engineered cells shown herein expressing the two glycosyltransferases (β-1,3-N-acetyl-glucosaminyltransferase and β-1,3-Galactosyltransferase) required for LNT synthesis and in addition the Smob enzyme produce almost exclusively LNFP-I.
(78) In one or more exemplary embodiments, the LNFP-I molar % fraction in the final HMO blend is more than 70%, such as but not limited to, more than 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.5%.
(79) In this manner, the present disclosure demonstrates an *E. coli* strain, that already produces LNT, can be advantageously employed to either increase the LNFP-I content of neutral HMO blends or establish an in vivo production process that results in an almost “pure” LNFP-I HMO product or at least a high-ratio LNFP-I:HMO by-product.
(80) Thus, in one or more exemplary embodiments, the methods described herein relates to the production of one or more HMOs, wherein the harvested HMOs contain 80-99.9%, preferably 90-99.9% LNFP-I.
(81) Performance Evaluation
(82) In one or more exemplary embodiments, the performance of the engineered HMO producing strains are evaluated two-fold. Firstly, overall productivity is assessed by the % change in HMO titer of one or more given HMO(s), measured as concentrations under comparable culture conditions. Secondly, quality of a given product composition was assessed by the ratio between a specific target HMO and another specific HMO by-products or the ratio between the target HMO and the sum of all HMOs. HMO ratios are in all instances given as molar ratios, either in the form of an absolute ratio as x:y or as a relative ratio given in %. For the purpose of the present invention lactose is not considered an HMO.
(83) Other HMOs
(84) 2′-FL
(85) 2′-Fucosyllactose (2′-FL or 2′O-fucosyllactose) is a trisaccharide, more precisely, fucosylated, neutral trisaccharide composed of L-fucose, D-galactose, and D-glucose units (Fucα1-2Galβ1-4Glc). It is the most prevalent human milk oligosaccharide (HMO) naturally present in human breast milk, making up about 30% of all of HMOs. In a genetically engineered cell or in an enzymatic reaction, 2′-FL is produced primarily by an α-1,2-fucosyltranferase enzymatic reaction with lactose and a fucosyl donor.
(86) In one or more exemplary embodiments, a HMO produced by the methods described herein is 2′-Fucosyllactose.
(87) LNT-II
(88) Lacto-N-triose II (LNT-II) is a trisaccharide, more precisely, a neutral trisaccharide composed of N-acetylglucosamine, galactose, and glucose (GlcNAcβ1-3Galβ1-4Glc). It is naturally present in human milk.
(89) In one or more exemplary embodiments, a HMO produced by the methods described herein is

Lacto-N-triose II.
(90) LNT
(91) Lacto-N-tetraose (LNT) is a tetrasaccharide, more precisely, a neutral tetrasaccharide composed of galactose, N-acetylglucosamine, galactose, and glucose (Galβ1-3GlcNAcβ1-3Galβ1-4Glc). It is naturally present in human milk.
(92) In one or more exemplary embodiments, a HMO produced by the methods described herein is Lacto-N-tetraose.
(93) In one or more exemplary embodiments, the method may produce one or more HMOs, wherein the predominant HMO produced is LNT.
(94) DFL
(95) Difucosyllactose (DFL or 2′,3-di-O-fucosyllactose) is tetrasaccharide, more precisely a fucosylated neutral tetrasaccharide composed of L-fucose, D-galactose, L-fucose, and D-glucose (Fucα1-2GalB1-4(Fucα1-3)Glc). It is naturally present in human milk.
(96) HMO Blend Molar Ratios
(97) The HMO products produced by the methods disclosed herein can also be given in ratios. The "ratio" as described herein is understood as the molar ratio between two HMOs or between one HMO and the sum of other HMOs. HMO ratios are in all instances given as molar ratios, either in the form of an absolute ratio as x:y or as a relative ratio given in %.
(98) Thus, in one or more exemplary embodiments, the molar ratio of LNFP-I:2′-FL harvested in step c) is in the range of 20:1-1:1, such as but not limited to 15:1, 10:1, 5:1, 1:1 or 2′-FL is absent from said product.
(99) In one or more exemplary embodiments, the molar ratio of LNFP-I:LNT harvested in step c) is in the range of 1000:1 to 1:1, such as but not limited to 100:1, 80:1, 50:1, 25:1, 15:1, 10:1, 5:1, 1:1 or LNT is absent from said product.
(100) The Molar Ratio of LNFP-1:2′-FL in the Harvested HMOs
(101) In one or more exemplary embodiments, the molar ratio of LNFP-I:2′-FL in the harvested HMOs in step c) according to the method, or in the final product is in the range of 15:1-2:1.
(102) In one or more exemplary embodiments, the molar ratio of LNFP-I:2′-FL in the harvested HMOs in step c) according to the method, or in the final product is in the range of 15:1-3:1, such as in the range 14:1 to 7:1, such as in the range 13:1 to 9:1.
(103) In one or more exemplary embodiments, the molar ratio of LNFP-I:2′-FL in the harvested HMOs in step c) according to the method, is in the range of 15:1 to 10:1, such as in the range 14:1 to 11:1, such as in the range 13:1 to 12:1.
(104) In one or more exemplary embodiments, the molar ratio of LNFP-I:2′-FL in the harvested HMOs in step c) according to the method, is 10:1, 5:1, or 3:1.
(105) The Ratio of LNFP-I:LNT in the Harvested HMOs
(106) In one or more exemplary embodiments, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method is in the range of 1000:1 to 10:1.
(107) In one or more exemplary embodiments, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method is in the range of 500:1 to 100:1.
(108) In one or more exemplary embodiments, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method is in the range of 800:1 to 200:1.
(109) In one or more exemplary embodiments, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method is in the range of 1000:1 to 500:1.
(110) In one or more exemplary embodiments, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method, is in the range of 950:1 to 750:1.
(111) In an exemplified embodiment, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method of the invention, is 100:1, 80:1, 50:1, or 10:1.
(112) In another exemplified embodiment, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method of the invention, is in the range of 90:1 to 11:1.

(113) In another exemplified embodiment, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method of the invention, is in the range of 50:1 to 20:1.
(114) In another exemplified embodiment, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method of the invention, is above 10:1, but below 100:1, such as above 20:1, but below 90:1 or such as above 50:1, but below 99:1.
(115) In another exemplified embodiment, the molar ratio of LNFP-I:LNT in the harvested HMOs in step c) according to the method of the invention, is above 10:1, but below 100:1, such as above 15:1, but below 50:1 or such as above 20:1, but below 40:1.
(116) Absent HMOs
(117) In one or more exemplary embodiments, LNT-II is absent from the harvested HMOs.
(118) An HMO is considered absent from the harvested HMOs in step c) in the methods described herein, when the amount of said HMO constitute less than 1% of the total amount of harvested HMOs, such as but not limited to less than 0.9%, less than 0.1%, less than 0.01%, less than 0.001% of the total amount of harvested HMOs.
(119) In one or more exemplary embodiments, 2′-FL and/or LNT-II and/or LNT and/or LNDFH-I and/or DFL is absent.
(120) In one or more exemplary embodiments, LNT-II and/or LNT is absent.
(121) In one or more exemplary embodiments, 2′-FL and/or LNDFH-I and/or DFL is absent.
(122) Culturing
(123) In the present context, culturing refers to the process by which cells are grown under controlled conditions, generally outside their natural environment, thus a method used to cultivate, propagate and grow a large number of cells.
(124) Cell Culture Medium
(125) In the present context, a growth medium or culture medium is a liquid or gel designed to support the growth of microorganisms, cells, or small plants. The medium comprises an appropriate source of energy and may comprise compounds which regulate the cell cycle. The culture medium may be semi-defined, i.e. containing complex media compounds (e.g. yeast extract, soy peptone, casamino acids, etc.), or it may be chemically defined, without any complex compounds. Exemplary suitable media are provided in experimental examples.
(126) In one or more exemplary embodiments, the culture media is minimal media.
(127) In one or more exemplary embodiments, the culture media is supplemented with one or more energy and carbon sources selected form the group containing lactose, glycerol, sucrose, glucose and fructose.
(128) In one or more exemplary embodiments, the culturing media is supplemented with one or more energy and carbon sources selected form the group containing glycerol, sucrose and glucose.
(129) In one or more exemplary embodiments, the culturing media is supplemented with glycerol, sucrose and/or glucose.
(130) In one or more exemplary embodiments, the culturing media is supplemented with glycerol and/or glucose.
(131) In one or more exemplary embodiments, the culturing media is supplemented with sucrose and/or glucose.
(132) In one or more exemplary embodiments, the culturing media is supplemented with glycerol and/or sucrose.
(133) In one or more exemplary embodiments, the culturing media is supplemented only with sucrose.
(134) In one or more exemplary embodiments, the culturing media contains sucrose as the sole carbon and energy source.
(135) In one or more exemplary embodiments, the culturing media is supplemented only with glucose.
(136) In one or more exemplary embodiments, the culturing media contains glucose as the sole

carbon and energy source.
(137) Sucrose Fermentation
(138) Biotechnological industry strives to develop (an) aerobic bioprocesses fueled by abundant and cheap carbon sources, like sucrose, thus in one or more exemplary embodiments, the genetically engineered cell is capable of utilizing sucrose as sole carbon and energy source.
(139) In one or more preferred exemplary embodiment(s), the genetically engineered cell expresses a sucrose utilization system. Such a system can be endogenous to the cell, but it may also be heterologous if the cell is not capable of utilizing sucrose.
(140) In one or more preferred exemplary embodiment(s), the genetically engineered cell comprises one or more heterologous nucleic acid sequence encoding one or more heterologous polypeptide(s), which enables utilization of sucrose as sole carbon and energy source of said genetically engineered cell.
(141) In one or more exemplary embodiments, the genetically engineered cell may comprise a PTS-dependent sucrose utilization transport system and/or a recombinant nucleic acid sequence encoding a heterologous polypeptide capable of hydrolysing sucrose into fructose and glucose.
(142) Such cells are capable of utilizing sucrose as carbon and energy source. For example, the culturing step according to step b) of the method(s) disclosed herein comprises a two-step sucrose feeding, with a second feeding phase by continuously adding to the culture an amount of sucrose that is less than that added continuously in a first feeding phase so as to slow the cell growth and increase the content of product produced in the high cell density culture.
(143) The feeding rate of sucrose added continuously to the cell culture during the second feeding phase may be around 30-40% less than that of sucrose added continuously during the first feeding phase.
(144) During both feeding phases, lactose can be added continuously, preferably with sucrose in the same feeding solution, or sequentially.
(145) Optionally, the culturing further comprises a third feeding phase when considerable amount of unused acceptor remained after the second phase in the extracellular fraction.
(146) Then the addition of sucrose is continued without adding the acceptor, preferably with around the same feeding rate set for the second feeding phase until consumption of the acceptor.
(147) In one or more exemplary embodiments, the genetically engineered cell may comprise one or more heterologous nucleic acid sequence encoding one or more heterologous polypeptide(s) which enables utilization of sucrose as sole carbon and energy source of said genetically engineered cell.
(148) In one or more exemplary embodiments, the genetically engineered cell expresses a sucrose utilization system selected from a PTS-dependent sucrose utilization system, further comprising the scrYA and scrBR operons. One PTS-dependent sucrose system is described in WO2015197082.
(149) In one or more exemplary embodiments, the polypeptide encoded by the scrYA operon are polypeptides with an amino acid sequence according to SEQ ID NOs: 9 and 10 [scrY and scrA] or a functional homologue of any one of SEQ ID NOs: 9 and 10 [scrY and scrA], having an amino acid sequence which is at least 80% identical to any one of SEQ ID NO: 9 and 10 [scrY and scrA].
(150) In one or more exemplary embodiments the polypeptide encoded by the scrBR operon are polypeptides with an amino acid sequence according to SEQ ID NOs: 11 and 12 [scrB and scrR] or a functional homologue of any one of SEQ ID NOs: 11 and 12 [scrB and scrR], having an amino acid sequence which is at least 80% identical to any one of SEQ ID NOs: 11 and 12 [scrB and scrR].
(151) In one or more preferred exemplary embodiment(s), the genetically engineered cell expresses a sucrose utilization system selected from a polypeptide capable of hydrolysing sucrose into glucose and fructose. Preferably, polypeptide capable of hydrolysing sucrose into glucose and fructose is a single heterologous enzyme.
(152) In one or more exemplary embodiments, the polypeptide capable of hydrolyzing sucrose into fructose and glucose is selected from the group consisting of SEQ ID NOs: 13 or 14 [SacC_Agal

and Bff], or a functional homologue of any one of SEQ ID NOs: 13 or 14 [SacC_Agal and Bff], having an amino acid sequence which is at least 80% identical, such as at least 85%, such as at least 90%, such as at least 95% identical to any one of SEQ ID NO: 13 or 14 [SacC_Agal and Bff]
(153) ScrB, ScrR, ScrY and ScrA
(154) ScrB is the heterologous polypeptide that is 100% identical to the GenBank ID: WP_000056853.1 or a functional homologue thereof having an amino acid sequence which is at least 80% identical and is a sucrose-6-phosphate hydrolase.
(155) ScrR is the heterologous polypeptide that is 100% identical to the GenBank ID: WP_000851062.1 or a functional homologue thereof having an amino acid sequence which is at least 80% identical and is a sucrose repressor protein.
(156) ScrY is the heterologous polypeptide that is 100% identical to the GenBank ID: CAA40657.1 or a functional homologue thereof having an amino acid sequence which is at least 80% identical and is a sucrose porin.
(157) ScrA is the heterologous polypeptide that is 100% identical to the GenBank ID: CAA40658.1 or a functional homologue thereof having an amino acid sequence which is at least 80% identical and is a sucrose-specific enzyme II.
(158) In the present context a sucrose utilization system is a group of heterologous polypeptides enabling the import and hydrolysis of sucrose into fructose and glucose as well as the DNA-level regulation of the system itself.
(159) Bff or SacC_AgaI
(160) SacC_AagI is the heterologous polypeptide which is 100% identical to the GenBank ID: WP_103853210.1 or a functional homologue thereof having an amino acid sequence which is at least 80% identical, such as at least 85%, such as at least 90%, such as at least 95% identical and is characterized as a glycoside hydrolase, and according to the current disclosure functions as an invertase and/or sucrose hydrolase.
(161) Bff is the heterologous polypeptide that is 100% identical to the GenBank ID: BAD18121.1 or a functional homologue, having an amino acid sequence which is at least 80% identical, such as at least 85%, such as at least 90%, such as at least 95% identical and is a β-fructofuranosidase.
(162) In the present context an invertase or sucrose hydrolase is an enzyme capable of hydrolysing sucrose into fructose and glucose.
(163) Harvesting
(164) The term “harvesting” in the context relates to collecting the produced HMO(s) following the termination of fermentation. In one or more exemplary embodiments it may include collecting the HMO(s) included in both the biomass (i.e. the host cells) and cultivation media, i.e. before/without separation of the fermentation broth from the biomass. In other embodiments, the produced HMOs may be collected separately from the biomass and fermentation broth, i.e. after/following the separation of biomass from cultivation media (i.e. fermentation broth). The definition of step c) is post fermentation.
(165) The separation of cells from the medium can be carried out with any of the methods well known to the skilled person in the art, such as any suitable type of centrifugation or filtration. The separation of cells from the medium can follow immediately after harvesting the fermentation broth or be carried out at a later stage after storing the fermentation broth at appropriate conditions. Recovery of the produced HMO(s) from the remaining biomass (or total fermentation) include extraction thereof from the biomass (i.e the production cells). It can be done by any suitable methods of the art, e.g. by sonication, boiling/heating, homogenization, enzymatic lysis using lysozyme, or freezing and grinding.
(166) After recovery from fermentation, HMO(s) are available for further processing and purification.
(167) Genetically Engineered Cell
(168) The present disclosure relates to a genetically engineered cell comprising a recombinant

nucleic acid sequence encoding a smob α-1,2-fucosyltransferase protein as described above. The present disclosure relates further to a genetically engineered cell for use in a method for producing fucosylated oligosaccharides. Said genetically engineered cell has been genetically engineered to express a heterologous fucosyl-transferase, which is capable of transferring a fucose residue from a donor substrate to an acceptor molecule, wherein said acceptor molecule is preferably lacto-N-tetraose.

(169) A "genetically engineered cell" as used herein is understood as a cell which has been transformed or transfected, by a recombinant nucleic acid sequence. Accordingly, a "genetically engineered cell" is in the present context understood as a host cell which has been transformed or transfected by a recombinant nucleic acid sequence.

(170) An aspect of the present invention is a genetically engineered cell comprising a recombinant nucleic acid sequence encoding an α-1,2-fucosyltransferase protein as shown in SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 70%, such as at least 80%, such as at least 90% such as at least 95% identical to SEQ ID NO: 1.

(171) In a further embodiment of the present invention the genetically engineered cell is an LNT producing cell allowing production of LNFP-I from lactose as the substrate. Preferably, the genetically engineered cell further comprises a nucleic acid sequence encoding a β-1,3-N-acetyl-glucosaminyltransferase protein and/or a nucleic acid sequence encoding a β-1,3-galactosyltransferase protein.

(172) To reduce the formation of 2′-FL it may be preferred to increase the β-1,3-N-acetyl-glucosaminyltransferase activity in the cell compared to the fucosyltransferase activity to favour addition of N-acetyl-glucosamine (GlcNAc) to lactose over fucose.

(173) In a further embodiment, the β-1,3-N-acetyl-glucosaminyltransferase expression levels are increased either by increasing the copy number and/or by choosing a strong regulatory element to control the expression.

(174) In a further embodiment, the β-1,3-galactosyltransferase expression levels are increased either by increasing the copy number and/or by choosing a strong regulatory element to control the expression.

(175) In a further embodiment, both the β-1,3-N-acetyl-glucosaminyltransferase and the β-1,3-galactosyltransferase expression levels are increased either by increasing the copy number and/or by choosing a strong regulatory element to control the expression.

(176) In further embodiments of the invention the genetically engineered cell of the present invention may contain further modifications described in the sections below, such as regulatory elements, in particular promoters and activators, repressor deletions, functional enzymes, such as sugar efflux transporters, additional glycosyl transferases, colanic acid pathway modifications etc.

(177) In one or more exemplary embodiments, the cell is capable of producing one or more HMO(s) selected from the group consisting of 2′-FL, LNT-II, LNT, LNFP-I, LNDFH-I and DFL.

(178) In one or more exemplary embodiments, the genetically engineered cell is capable of producing one or more HMO(s) selected from the group consisting of 2′-FL, LNT-II, LNT and LNFP-I.

(179) In one or more exemplary embodiments, the predominant HMO produced by the genetically engineered cell is LNFP-I.

(180) The genetically engineered cell may be any cell useful for HMO production including mammalian cell lines. Preferably, the host cell is a unicellular microorganism of eucaryotic or prokaryotic origin.

(181) Appropriate microbial cells that may function as a host cell include yeast cells, bacterial cells, archaebacterial cells, algae cells, and fungal cells.

(182) The genetically engineered cell (host cell) may be e.g. a bacterial or yeast cell. In one preferred embodiment, the genetically engineered cell is a prokaryotic cell, such as a bacterial cell.

(183) Bacterial Host Cells

(184) Regarding the bacterial host cells, there are, in principle, no limitations; they may be eubacteria (gram-positive or gram-negative) or archaebacteria, as long as they allow genetic manipulation for insertion of a gene of interest and can be cultivated on a manufacturing scale. Preferably, the host cell has the property to allow cultivation to high cell densities. Non-limiting examples of bacterial host cells that are suitable for recombinant industrial production of an HMO(s) according to the invention could be *Erwinia herbicola* (*Pantoea agglomerans*), *Citrobacter freundii, Pantoea citrea, Pectobacterium carotovorum*, or *Xanthomonas campestris*. Bacteria of the genus Bacillus may also be used, including *Bacillus subtilis, Bacillus licheniformis, Bacillus coagulans, Bacillus thermophilus, Bacillus laterosporus, Bacillus megaterium, Bacillus mycoides, Bacillus pumilus, Bacillus lentus, Bacillus cereus, and Bacillus circulans*. Similarly, bacteria of the genera *Lactobacillus* and *Lactococcus* may be engineered using the methods of this invention, including but not limited to *Lactobacillus acidophilus, Lactobacillus salivarius, Lactobacillus plantarum, Lactobacillus helveticus, Lactobacillus delbrueckii, Lactobacillus rhamnosus, Lactobacillus bulgaricus, Lactobacillus crispatus*, *Lactobacillus gasseri*, *Lactobacillus casei*, *Lactobacillus reuteri*, *Lactobacillus jensenii*, and *Lactococcus lactis. Streptococcus thermophiles* and *Proprionibacterium freudenreichii* are also suitable bacterial species for the invention described herein. Also included as part of this invention are strains, engineered as described here, from the genera *Enterococcus* (e.g., *Enterococcus faecium* and *Enterococcus thermophiles*), *Bifidobacterium* (e.g., *Bifidobacterium longum, Bifidobacterium infantis*, and *Bifidobacterium bifidum*), *Sporolactobacillus* spp., *Micromomospora* spp., *Micrococcus* spp., *Rhodococcus* spp., and *Pseudomonas* (e.g., *Pseudomonas fluorescens* and *Pseudomonas aeruginosa*).
(185) Non-limiting examples of fungal host cells that are suitable for recombinant industrial production of an HMO(s) according to the invention could be yeast cells, such as *Komagataella phaffii*, *Kluyveromyces lactis*, *Yarrowia lipolytica*, *Pichia pastoris*, and *Saccaromyces cerevisiae* or filamentous fungi such as *Aspargillus* sp, *Fusarium* sp or *Thricoderma* sp, exemplary species are *A. niger, A. nidulans, A. oryzae, F. solani, F. graminearum* and *T. reesei.*
(186) In one or more exemplary embodiments, the genetically engineered cell is *S. cerevisiae* or *P pastoris.*
(187) In one or more exemplary embodiments, the genetically engineered cell is *Pichia pastoris.*
(188) In one or more exemplary embodiments, the genetically engineered cell is *S. cerevisiae.*
(189) In one or more exemplary embodiments, the genetically engineered cell is selected from the group consisting of *E. coli, C. glutamicum, L. lactis, B. subtilis, S. lividans, P. pastoris*, and *S. cerevisiae.*
(190) In one or more exemplary embodiments, the genetically engineered cell is selected from the group consisting of *B. subtilis, S. cerevisiae* and *Escherichia coli.*
(191) In one or more exemplary embodiments, the genetically engineered cell is *B. subtilis.*
(192) In one or more exemplary embodiments, the genetically engineered cell is *Escherichia coli.*
(193) In one or more exemplary embodiments, the invention relates to a genetically engineered cell, wherein the cell is derived from the *E. coli* K-12 strain or DE3.
(194) Controlling the Expression
(195) In the present context the term “controlling the expression” relates to gene expression where the transcription of a gene into mRNA and its subsequent translation into protein is controlled. Gene expression is primarily controlled at the level of transcription, largely as a result of binding of proteins to specific sites on DNA, such as but not limited to regulatory elements.
(196) As described above, engineering strategy can be applied in multiple ways: 1) the copy number 2) the controlling the expression of any copy of these genes at the transcriptional or the translational level 3) the deletion of regulators that repress the expression of key genes in the HMO production process 4) the over-expression of regulators that activate and/or enhance the expression of key genes in the HMO production process

(197) Increasing the gene copy number and/or the expression of genes coding the enzymes that are directly involved in the LNFP-I and 2′-FL biosynthetic pathways, including the synthesis of the activated sugars GDP-fucose, UDP-N-acetyl-glucosaminyl and UDP-Gal (donor sugars) and the decoration of lactose, LNT-II and LNT (acceptor sugars) to form, respectively, LNT-II or 2′-FL, LNT, and LNFP-I is desired, in particular LNFP-I.
(198) In one or more exemplary embodiments, the expression of different heterologous and/or native genes I the cell are controlled to achieve the optimal balance increasing either total HMO yield and/or the molar % of the total HMO of one or more selected HMOs, such as LNFP-I
(199) Over-Expression
(200) A variety of molecular mechanisms ensures that genes are expressed at the appropriate level and under conditions of relevance to the applied production process. For instance, the regulation of transcription can be summarized into the following routes of influence; genetic (direct interaction of a control factor with the gene of interest), modulation and/or interaction of a control factor within the transcriptional machinery and epigenetic (non-sequence changes in DNA structure that influence transcription).
(201) It is known that a reduction in gene expression below a critical threshold for any gene will result in a mutant phenotype, since such a defect essentially mimics either a partial or complete loss of function of the target gene, whereas increased expression of a native gene can be both beneficial or disruptive to a cell or organism.
(202) Over-expression of a gene may be achieved directly by transcriptional activators that bind to key gene regulatory sequences to promote transcription or enhancers that constitute sequence elements positively affecting transcription. Similarly, direct over-expression of a gene can be achieved by simply increasing its copy number in the genome, or replacing its native promoter with a promoter of higher strength or even modifying the sequence controlling the binding of the corresponding mRNA to the ribosomes, i.e. the Shine-Dalgarno sequence being present upstream of the gene's coding sequence.
(203) Moreover, over-expression of a gene may also be achieved indirectly through the partial or full inactivation of transcriptional repressors that normally bind key regulatory sequences around the coding sequence of the gene of interest and thereby inhibit its transcription.
(204) Thus, in one or more exemplary embodiments, the over-expression of the protein(s) can be is provided by increasing the copy number of the genes coding said protein(s), and/or by choosing an appropriate element for or adding an extra genomic copy, and/or conferring a non-functional (or absent) gene product that normally binds to and repress the expression.
(205) Increasing the Copy Number
(206) Copy number variation is a type of structural variation: specifically, it is a type of duplication or multiplication of a considerable number of base pairs which if representing a protein encoding gene will result in an increase of the number of genes encoding the same protein. Such variation can occur naturally in many species but can also be introduced by genetically modifying a host cell
(207) In one or more exemplary embodiments, expression is controlled by increasing the copy number of the desired genes. Copy numbers can be increased either by introducing a plasmid which has a high copy number in the cell or by introducing an additional copy of the gene into the genome of the host cell.
(208) Thus, in one or more exemplary embodiments, the present disclosure relates to a method, wherein the overexpression of the protein(s) is provided by increasing the copy number of the genes coding for said protein(s) and/or by choosing an appropriate regulatory element.
(209) Regulatory Element
(210) The genetically engineered cell may comprise recombinant genes and/or nucleic acids of homologous or heterologous origin. The expression of said recombinant genes and/or nucleic acids can be regulated by one or more nucleic acid sequences comprising a regulatory element.
(211) The term “regulatory element” is to be understood as a regulatory nucleic acid sequence that

modulates the expression of a nucleic acid sequence comprising e.g., a coding sequence.
(212) In general, the transcriptional and translational regulatory sequences include, but are not limited to, promoter sequences, ribosomal binding sites, transcriptional start and stop sequences, translational start and stop sequences, and enhancer or activator sequences.
(213) In one or more exemplary embodiments, the expression of the recombinant nucleic acid is regulated by one or more nucleic acid sequences comprising a regulatory element.
(214) In one or more exemplary embodiments, the regulatory element or elements comprise(s) a promoter sequence.
(215) In one or more exemplary embodiments, the recombinant regulatory element comprises more than one promoter sequence, or a recombinant promoter sequence with elements combined from different promoters.
(216) In one or more exemplary embodiments, the recombinant regulatory element comprises a single promoter sequence, suitable for regulating the genes and/or heterologous nucleic acid sequences of the genetically engineered cell of the present invention.
(217) In one or more exemplary embodiments, the recombinant regulatory element comprises two or more regulatory elements with identical promoter sequences suitable for regulating the genes and/or heterologous nucleic acid sequences of the genetically engineered cell of the present invention.
(218) In one or more exemplary embodiments, the recombinant regulatory element comprises two or more regulatory elements with non-identical promoter sequences, suitable for regulating the genes and/or heterologous nucleic acid sequences of the genetically engineered cell of the present invention.
(219) The regulatory architectures i.e., gene-by-gene distributions of transcription-factor-binding sites and identities of the transcription factors that bind those sites can be used multiple different growth conditions and there are more than 100 genes from across the *E. coli* genome, which acts as regulatory elements. Thus, any promoter sequence enabling transcription and/or regulation of the level of transcription, of one or more heterologous or native nucleic acid sequences that encode one or more proteins as described herein may be suitable.
(220) Promoters
(221) The regulatory element or elements, as described above, regulating the expression of the genes and/or nucleic acid sequence(s), may comprise one or more promoter sequence(s), wherein the promoter sequence, is operably linked to the nucleic acid sequence of the gene of interest in that sense regulating the expression of the nucleic acid sequence of the gene of interest.
(222) The term “operably linked” as used herein, refers to a functional relationship between two or more nucleic acid (e.g., DNA) segments. Operably linked refers to the functional relationship of a transcriptional regulatory sequence (such as a promoter sequence, signal sequence, or array of transcription factor binding sites) to a transcribed sequence. For example, a promoter sequence is operably linked to a coding sequence if it stimulates or modulates the transcription of the coding sequence in an appropriate host cell or other expression system. Accordingly, the term “promoter sequence” designates DNA sequences which usually “precede” a gene in a DNA polymer and provide a site for initiation of the transcription into mRNA. “Regulator” DNA sequences, also usually “upstream” of (i.e., preceding) a gene in a given DNA polymer, bind proteins that determine the frequency (or rate) of transcriptional initiation. Collectively referred to as “promoter/regulator sequence” or “control” DNA sequence, these sequences which precede a selected gene (or series of genes) in a functional DNA polymer cooperate to determine whether the transcription (and eventual expression) of a gene will occur. DNA sequences which “follow” a gene in a DNA polymer and provide a signal for termination of the transcription into mRNA are referred to as transcription “terminator” sequences.
(223) Generally, promoter sequences that are operably linked to a transcribed sequence are physically contiguous to the transcribed sequence, i.e., they are cis-acting.

(224) The regulatory element, such as a promoter, controls the expression of the mentioned glycosyltransferases, transporters and the colanic acid gene cluster, and this regulatory element should precede the coding sequence of the construct (promoter/regulatory element+coding sequence). The construct may be integrated into the genome, or it can be introduced into the cell in the form of a plasmid or another episomal element.
(225) The promoter may be of heterologous origin, native to the genetically modified cell or it may be a recombinant promoter, combining heterologous and/or native elements. In general, a promoter may comprise native, heterologous and/or synthetic nucleic acid sequences, and may be a recombinant nucleic acid sequence, recombining two or more nucleic acid sequences or same or different origin as described above, thereby generating a homologous, heterologous or synthetic nucleic promoter sequence, and/or a homologous, heterologous or synthetic nucleic regulatory element.
(226) One way to increase the production of a product may be to regulate the production of the desired enzyme activity used to produce the product, such as the glycosyltransferases, transporters or enzymes involved in the biosynthetic pathway of the glycosyl donor.
(227) Increasing the promoter strength driving the expression of the desired enzyme may be one way of doing this. The strength of a promoter can be assed using a lacZ enzyme assay where β-galactosidase activity is assayed as described previously (see e.g. Miller J. H. *Experiments in molecular genetics*, Cold spring Harbor Laboratory Press, NY, 1972). Briefly the cells are diluted in Z-buffer and permeabilized with sodium dodecyl sulfate (0.1%) and chloroform. The LacZ assay is performed at 30° C. Samples are preheated, the assay initiated by addition of 200 μl ortho-nitro-phenyl-β-galactosidase (4 mg/ml) and stopped by addition of 500 μl of 1 M Na.sub.2CO.sub.3 when the sample had turned slightly yellow. The release of ortho-nitrophenol is subsequently determined as the change in optical density at 420 nm. The specific activities are reported in Miller Units (MU) [A420/(min*ml*A600)]. A regulatory element with an activity above 10,000 MU is considered strong and a regulatory element with an activity below 3,000 MU is considered weak, what is in between has intermediate strength. An example of a strong regulatory element is the PglpF promoter with an activity of approximately 14.000 MU and an example of a weak promoter is Plac which when induced with IPTG has an activity of approximately 2300 MU. Table 4 below illustrates various suitable promoters of the invention sorted according to strength relative to the PglpF promoter.
(228) Alternatively, if there is a need for balancing the expression level of one or more proteins to optimize the production it may be beneficial to use a promoter with the desired strength, e.g., middle or low strength.
(229) TABLE-US-00001 TABLE 4 Selected promoter sequences % Activity relative Seq ID Promoter name to PglpF* Strength Reference in appl. PmglB_70UTR_SD8 291% high WO2020255054 15 PmglB_70UTR_SD10 233-281% high WO2020255054 16 PmglB_54UTR 197% high WO2020255054 17 Plac_70UTR 182-220% high WO2019123324 18 PmglB_70UTR_SD9 180-226% high WO2020255054 19 PmglB_70UTR_SD4 153%-353% high WO2020255054 20 PmglB_70UTR_SD5 146-152% high WO2020255054 21 PglpF_SD4 140-161% high WO2019123324 22 PmglB_70UTR_SD7 127-173% high WO2019123324 23 PmglB_70UTR 124-234% high WO2020255054 24 PglpA_70UTR 102-179% high WO2019123324 25 PglpT_70UTR 102-240% high WO2019123324 26 PglpF 100% high WO2019123324 27 PglpF_SD10 88-96% high WO2019123324 28 PglpF_SD5 82-91% high WO2019123324 29 PglpF_SD8 81-82% high WO2019123324 30 PmglB_16UTR 78-171% high WO2019123324 31 PglpF_SD9 73-93% middle WO2019123324 32 PglpF_SD7 47-57% middle WO2019123324 33 PglpF_SD6 46-47% middle WO2019123324 34 PglpA_16UTR 38-64% middle WO2019123324 35 Plac_wt* 15-28% low WO2019123324 36 PglpF_SD3 9% low WO2019123324 37 PglpF_SD1 5% low WO2019123324 38 PglpF_B28 Not 39 assessed *The promoter activity is assessed in the LacZ assay described below with the PglpF

promoter run as positive reference in the same assay. To compare across assays the activity is calculated relative to the PglpF promoter, a range indicates results from multiple assays
(230) In embodiments of the invention the expression of selected nucleic acid sequences of the present invention is under control of a PglpF (SEQ ID NO: 27) or Plac (SEQ ID NO: 36) or PmglB_UTR70 (SEQ ID NO: 24) or PglpA_70UTR (SEQ ID NO: 25) or PglpT_70UTR (SEQ ID NO: 26) or variants of these promoters as identified in Table 5.
(231) Specific PglpF variants can be selected from the group consisting of SEQ ID NO: 22, 28, 29, 30, 32, 33 or 34. A specific Plac variant is SEQ ID NO: 28. Specific PmglB_70UTR variants can be selected from the group consisting of SEQ ID NO: 15, 16, 17, 19, 20, 21, 23 or 31.
(232) Further suitable variants of PglpF, PglpA_70UTR, PglpT_70UTR and PmglB_70UTR promoter sequences are described in or WO2019/123324 and WO2020/255054 respectively (hereby incorporated by reference).
(233) In one or more exemplary embodiments, the regulatory element is a promoter with high or middle strength, such as a promoter sequence selected from the group consisting of PmglB_70UTR_SD8, PmglB_70UTR_SD10, PmglB_54UTR, Plac_70UTR, PmglB_70UTR_SD9, PmglB_70UTR_SD4, PmglB_70UTR_SD5, PglpF_SD4, PmglB_70UTR_SD7, PmglB_70UTR, PglpA_70UTR, PglpT_70UTR, pgatY_70UTR, PglpF, PglpF_SD10, PglpF_SD5, PglpF_SD8, PglpF_B28, PglpF_B29, PmglB_16UTR, PglpF_SD9, PglpF_SD7, PglpF_SD6 and PglpA_16UTR
(234) In on referred embodiment the promoter is a strong promoter selected from the group consisting of PmglB_70UTR_SD8, PmglB_70UTR_SD10, PmglB_54UTR, Plac_70UTR, PmglB_70UTR_SD9, PmglB_70UTR_SD4, PmglB_70UTR_SD5, PglpF_SD4, PmglB_70UTR_SD7, PmglB_70UTR, PglpA_70UTR, PglpT_70UTR, pgatY_70UTR, PglpF, PglpF_SD10, PglpF_SD5, PglpF_SD8, and PmglB_16UTR. This may in particular be advantageous for the expression the glycosyltransferases.
(235) In another embodiment the promoter is selected from the group consisting of promoters with middle strength, such as PglpF_SD9, PglpF_SD7, PglpF_SD6 and PglpA_16UTR.
(236) In another embodiment the promoter is selected from the group consisting of promoters with low strength, such as Plac_wt. PglpF_SD3 and PglpF_SD1.
(237) In one or more exemplary embodiments, the regulatory element for the regulation of the expression of a recombinant nucleic acid sequence construct of the present disclosure is the glpFKX operon promoter sequence, PglpF.
(238) In a presently preferred embodiment, the promoter sequence is PglpF.
(239) In one or more exemplary embodiments, the promoter sequence is selected from the group consisting of PBAD, Ptet, Pxyl, PsacB, PxylA, PrpR, PnitA, PT7, Ptac, PL, PR, PnisA, Pb, PgatY_70UTR, PglpF, PglpF_SD1, PglpF_SD10, PglpF_SD2, PglpF_SD3, PglpF_SD4, PglpF_SD5, PglpF_SD6, PglpF_SD7, PglpF_SD8, PglpF_SD9, PglpF_B28, Plac_16UTR, Plac, PmglB_70UTR and PmglB_70UTR_SD4.
(240) In one or more exemplary embodiments, the promoter sequence is selected from the group consisting of PglpF, PglpF_SD1, PglpF_SD10, PglpF_SD2, PglpF_SD3, PglpF_SD4, PglpF_SD5, PglpF_SD6, PglpF_SD7, PglpF_SD8, PglpF_SD9 and PglpF_B28.
(241) In one or more exemplary embodiments, the promoter sequence is selected from the group consisting of PglpF and PglpF_B28. PglpF_B28 is an engineered version of the PglpF sequence comprising an engineered ribosomal binding site sequence downstream of the promoter sequence.
(242) Apart from nucleic acid(s) encoding glycosyltransferase(s), the coding sequence may also comprise one or more recombinant nucleic acid(s) encoding one or more gene regulatory proteins and/or metabolic enzymes of the production host organism and/or a native or heterologous sugar efflux transporter and/or proteins enabling the utilization of sucrose as a carbon and energy source.
(243) Repressors
(244) In one or more exemplary embodiment(s), the genetically engineered cell disclosed herein comprises a non-functional or absent gene product that normally binds to a regulatory element and

represses the expression of any of the proteins of the present disclosure regulated by said regulatory element.
(245) The term a non-functional (or absent) gene product that normally binds to and represses the expression driven by the regulatory element in the present context relates to DNA binding sites upstream of the coding sequence of a gene of interest and specifically at the promoter region of said gene.
(246) In one or more exemplary embodiments, the cell may have a non-functional (or absent) gene product(s) that would normally bind to and repress the expression of the α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1 or regions upstream of the regulatory element for controlling the expression of the α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1.
(247) In one or more exemplary embodiments, said gene product is the DNA-binding transcriptional repressor GlpR.
(248) GlpR
(249) GlpR belongs to the DeoR family of transcriptional regulators and acts as the repressor of the glycerol-3-phosphate regulon, which is organized in different operons. This regulator is part of the glpEGR operon, yet it can also be constitutively expressed as an independent (glpR) transcription unit. In addition, the operons regulated are induced when *Escherichia coli* is grown in the presence of inductor, glycerol, or glycerol-3-phosphate (G3P), and the absence of glucose. In the absence of inductor, this repressor binds in tandem to inverted repeat sequences that consist of 20-nucleic acid-long DNA target sites.
(250) The term "non-functional or absent" in relation to the glpR gene refers to the inactivation of the glpR gene by complete or partial deletion of the corresponding nucleic acid sequence from the bacterial genome (e.g. SEQ ID NO: 42 or variants thereof encoding glpR capable of downregulating glpF derived promoters). The glpR gene encodes the DNA-binding transcriptional repressor GlpR. In this way promoter sequences of the PglpF family are more active in the genetically engineered cell, due to deletion of the repressor gene that would otherwise reduce the transcriptional activity associated with the PglpF promoters.
(251) In one or more exemplary embodiments, the glpR gene is deleted.
(252) The deletion of the glpR gene could eliminate the GlpR-imposed repression of transcription from all PglpF promoters in the cell and in this manner enhance gene expression from all PglpF-based cassettes.
(253) Activators
(254) In one or more exemplary embodiment(s), the genetically engineered cell disclosed herein comprises an over-expressed gene product that enhances the expression of any of the proteins of the present disclosure regulated by said regulatory element.
(255) In one or more exemplary embodiments, the cell of the present disclosure may comprise an over-expressed gene product that enhances the expression of the α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1 or regions upstream of the regulatory element for controlling the expression of the α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1.
(256) In one or more exemplary embodiments, said gene product is the cAMP DNA-binding transcriptional dual regulator CRP.
(257) CRP
(258) CRP belongs to the CRP-FNR superfamily of transcription factors. CRP regulates the expression of several of the *E. coli* genes, many of which are involved in catabolism of secondary carbon sources. Upon activation by cyclic-AMP, (cAMP) CRP binds directly to specific promoter sequences, the binding recruits the RNA polymerase through direct interaction, which in turn activates the transcription of the nucleic acid sequence following the promoter sequence leading to expression of the gene of interest. Thus, over-expression of CRP may lead to an enhanced expression of a gene/nucleic acid sequence of interest. Amongst other functions, CRP exerts its function on the PglpF promoters, where it contrary to the repressor GlpR, activates promoter

sequences of the PglpF family. In this way, over-expression of CRP in the genetically engineered cell of the present disclosure, promotes expression of genes that are regulated by promoters of the PglpF family.

(259) Thus, in one or more exemplary embodiments, the crp gene is over-expressed.

(260) Genetic engineering of GlpR and/or CRP, as suggested in the present disclosure, in 2′-FL producing strains is beneficial for the overall production of 2′-FL by these strains.

(261) Nucleic Acid Constructs

(262) An aspect of the present disclosure is the provision of a nucleic acid construct comprising a heterologous nucleic acid sequence(s) encoding an α-1,2-fucosyl-transferase protein as shown in SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 70% identical, such as at least 80%, such as at least 85%, such as at least 90%, such as at least 95% identical to SEQ ID NO: 1.

(263) In one or more exemplary embodiments, the nucleic acid construct comprising a recombinant nucleic acid sequence which is at least 70% identical to SEQ ID NO: 2, such as more than 80%, such as more than 85%, such as more than 90%, such as more than 95% or such as more than 99% identical to SEQ ID NO: 2.

(264) In an exemplified embodiment, the nucleic acid construct comprising a recombinant nucleic acid sequence which is identical to SEQ ID NO: 2.

(265) In one or more exemplary embodiments, the nucleic acid construct comprising a recombinant nucleic acid sequence or a functional homologue having a nucleic acid sequence which is at least 70% identical to any one of SEQ ID NO: 3, 4, 5, 6, 7, 8. SEQ ID NO: 3, 4, 5, 6, 7 and 8 encodes a polypeptide capable of transporting an HMO out of the cell. These constructs are functional or a functional homologue of any one of the GenBank accession IDs: WP_017489914.1, WP_092672081.1, EEQ08298.1, WP_087817556.1, WP_048785139.1 or WP_060448169.1 having an amino acid sequence which is at least 80% identical to any one of any one of the GenBank accession IDs: WP_017489914.1, WP_092672081.1, EEQ08298.1, WP_087817556.1, WP_048785139.1 or WP_060448169.1.

(266) In one or more exemplary embodiments the invention relates to a nucleic acid construct further comprising a recombinant nucleic acid sequence which is at least 70% identical, such as at least 80%, such as at least 85%, such as at least 90%, such as at least 95% identical to SEQ ID NO: 9 or 10 encoding a polypeptide capable of hydrolyzing sucrose into fructose and glucose of any one of the GenBank accession IDs: WP_103853210.1 and BAD18121.1, or a functional homologue of any one of the GenBank accession IDs: WP_103853210.1 and BAD18121.1, having an amino acid sequence which is at least 80% identical to any one of the GenBank accession IDs: WP_103853210.1 and BAD18121.1.

(267) The nucleic acid construct can be a recombinant nucleic acid sequence. By the term “recombinant nucleic acid sequence”, “recombinant gene/nucleic acid/DNA encoding” or “coding nucleic acid sequence” used interchangeably is meant an artificial nucleic acid sequence (i.e. produced in vitro using standard laboratory methods for making nucleic acid sequences) that comprises a set of consecutive, non-overlapping triplets (codons) which is transcribed into mRNA and translated into a protein when under the control of the appropriate control sequences, i.e. a promoter sequence.

(268) The boundaries of the coding sequence are generally determined by a ribosome binding site located just upstream of the open reading frame at the 5′end of the mRNA, a transcriptional start codon (AUG, GUG or UUG), and a translational stop codon (UAA, UGA or UAG). A coding sequence can include, but is not limited to, genomic DNA, cDNA, synthetic, and recombinant nucleic acid sequences.

(269) The term “nucleic acid” includes RNA, DNA and cDNA molecules. It is understood that, as a result of the degeneracy of the genetic code, a multitude of nucleic acid sequences encoding a given protein may be produced.

(270) In a presently preferred embodiment, the nucleic acid construct comprises a recombinant nucleic acid sequence which is identical to SEQ ID NO: 2, wherein the nucleic acid sequence encodes an α-1,2-fucosyltransferase of SEQ ID NO: 1.

(271) A Recombinant Nucleic Acid Sequence

(272) The recombinant nucleic sequence may be a coding DNA sequence e.g., a gene, or non-coding DNA sequence e.g., a regulatory DNA, such as a promoter sequence.

(273) Accordingly, in one exemplified embodiment the invention relates to a nucleic acid construct comprising a coding nucleic sequence, i.e. recombinant DNA sequence of a gene of interest, e.g. a fucosyltransferase gene, and a non-coding regulatory DNA sequence, e.g. a promoter DNA sequence, e.g. a recombinant promoter sequence derived from the promoter sequence of lac operon or an glp operon, or a promoter sequence derived from another genomic promoter DNA sequence, or a synthetic promoter sequence, wherein the coding and promoter sequences are operably linked.

(274) In one exemplified embodiment, the nucleic acid construct of the invention may be a part of the vector DNA, in another embodiment the construct it is an expression cassette/cartridge that is integrated in the genome of a host cell.

(275) Accordingly, the term “nucleic acid construct” means an artificially constructed segment of nucleic acid, in particular a DNA segment, which is intended to be ‘transplanted’ into a target cell, e.g. a bacterial cell, to modify expression of a gene of the genome or express a gene/coding DNA sequence which may be included in the construct.

(276) Integration of the nucleic acid construct of interest comprised in the construct (expression cassette) into the bacterial genome can be achieved by conventional methods, e.g. by using linear cartridges that contain flanking sequences homologous to a specific site on the chromosome, as described for the attTn7-site (Waddell C. S. and Craig N. L., Genes Dev. (1988) February; 2(2):137-49); methods for genomic integration of nucleic acid sequences in which recombination is mediated by the Red recombinase function of the phage λ or the RecE/RecT recombinase function of the Rac prophage (Murphy, J Bacteriol. (1998); 180(8):2063-7; Zhang et al., Nature Genetics (1998) 20:123-128 Muyrers et al., EMBO Rep. (2000) 1(3):239-243); methods based on Red/ET recombination (Wenzel et al., Chem Biol. (2005), 12(3):349-56; Vetcher et al., Appl Environ Microbiol. (2005); 71(4):1829-35); or positive clones, i.e. clones that carry the expression cassette, can be selected e.g. by means of a marker gene, or loss or gain of gene function.

(277) As described above, the disclosure enables the use of regulatory elements for the expression of a nucleic acid or gene of interest, and thus a nucleic acid construct may further comprise one or more recombinant nucleic acid sequence(s) comprising a regulatory element, such as a promoter sequence described in the section “promoters” above.

(278) As shown in the Examples and described above, particularly good results are achieved by using a nucleic acid construct, wherein said promoter sequence is PglpF.

(279) In another exemplified embodiment of the invention, the promoter sequence is lac operon promoter sequence, Plac.

(280) Functional Enzymes

(281) To be able to synthesize one or more HMOs, the recombinant cell of the present invention may further comprise the necessary functional enzyme with activity enabling a viable industrial process for HMO production.

(282) Sugar Efflux Transporter

(283) Over the past decade several new and efficient sugar efflux transporter proteins have been identified, each having specificity for different recombinantly produced HMOs and development of recombinant cells expressing said protein are advantageous for large-scale industrial HMO manufacturing. Sugar transport relates to the transport of a sugar, such as, but not limited to, an oligosaccharide.

(284) The genetically engineered cell(s) described herein, may also comprise a recombinant nucleic acid encoding a sugar efflux transporter. A sugar efflux transporter may for example enhance the

level of a HMO in a method as described herein. In one or more exemplary embodiments, the genetically engineered cell further comprises a gene product that acts as a sugar efflux transporter.
(285) Influx and/or efflux transport of one/or more HMOs, from the cytoplasm or periplasm of a genetically engineered cell as described herein to the production media and/or from the production media to the cytoplasm or periplasm is disclosed.
(286) A polypeptide, expressed in the genetically engineered cell as disclosed herein, capable of transporting one or more HMOs from the cytoplasm or periplasm to the production medium and/or from the production media to the cytoplasm or periplasm of a genetically engineered cell, is a polypeptide capable of sugar transport.
(287) Thus, in the present context, sugar transport can mean efflux and/or influx transport of sugar(s), such as, but not limited to, an HMO.
(288) Thus, in one or more exemplary embodiments, the genetically engineered cell according to the method described herein further comprises a gene product that acts as a sugar efflux transporter. The gene product that acts as a sugar efflux transporter may be encoded by a recombinant nucleic acid sequence that is expressed in the genetically engineered cell. The recombinant nucleic acid sequence encoding a sugar efflux transporter, may be integrated into the genome of the genetically engineered cell. It may be plasmid borne, or it may be part of an episomal expression element.
(289) MFS Transporters
(290) Exemplary sugar efflux transporters are a subspecies of the Major Facilitator Superfamily proteins. The MFS transporters facilitate the transport of molecules, such as but not limited to sugars like oligosaccharides, across the cellular membranes.
(291) By the term “Major Facilitator Superfamily (MFS)” is meant a large and exceptionally diverse family of the secondary active transporter class, which is responsible for transporting a range of different substrates, including sugars, drugs, hydrophobic molecules, peptides, organic ions, etc.
(292) The term “MFS transporter” in the present context means, a protein that facilitates transport of an oligosaccharide, preferably, an HMO, through the cell membrane, preferably transport of an HMO/oligosaccharide synthesized by the genetically engineered cell as described herein from the cell cytosol to the cell medium. Additionally, or alternatively, the MFS transporter may also facilitate efflux of molecules that are not considered HMO or oligosaccharides, such as lactose, glucose, cell metabolites and/or toxins.
(293) In one or more exemplary embodiments, the MFS transporter protein is selected from the group consisting of Bad, Nec, YberC, Fred, Vag and Marc.
(294) In one or more presently preferred exemplary embodiments, the sugar efflux transporter is Nec or YberC.
(295) Bad
(296) The MFS transporter protein identified herein as “Bad protein” or “Bad transporter” or “Bad”, interchangeably, has the amino acid sequence of SEQ ID NO: 3; The amino acid sequence identified herein as SEQ ID NO: 3 is an amino acid sequence that has 100% identity with the amino acid sequence having the GenBank accession ID WP_017489914.1.
(297) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is Bad.
(298) Nec
(299) The MFS transporter protein having the amino acid sequence of SEQ ID NO: 4 is identified herein as “Nec protein” or “Nec transporter” or “Nec”, interchangeably; a nucleic acid sequence encoding nec protein is identified herein as “Nec coding nucleic acid/DNA” or “nec gene” or “nec”; The amino acid sequence identified herein as SEQ ID NO: 4 is the amino acid sequence that is 100% identical to the amino acid sequence having the GenBank accession ID WP_092672081.1.
(300) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is Nec. In a further embodiment the sugar efflux transporter has the amino acid sequence of

SEQ ID NO: 4 or is a functional homologue having an amino acid sequence which is at least 70% identical, such as at least 80% identical, such as at least 85% identical, such as at least 90% identical, such as at least 95% identical or such as at least 99% identical to any one of SEQ ID NO: 4.
(301) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is Nec.
(302) YberC
(303) The MFS transporter protein having the amino acid sequence of SEQ ID NO: 5 is identified herein as “YberC protein” or “YberC transporter” or “YberC”, interchangeably; a nucleic acid sequence encoding yberC protein is identified herein as “YberC coding nucleic acid/DNA” or “yberC gene” or “yberC”; The amino acid sequence identified herein as SEQ ID NO: 5 is the amino acid sequence that is 100% identical to the amino acid sequence having the GenBank accession ID EEQ08298.1.
(304) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is YberC. In a further embodiment the sugar efflux transporter has the amino acid sequence of SEQ ID NO: 5 or is a functional homologue having an amino acid sequence which is at least 70% identical, such as at least 80% identical, such as at least 85% identical, such as at least 90% identical, such as at least 95% identical or such as at least 99% identical to any one of SEQ ID NO: 5.
(305) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is YberC.
(306) Fred
(307) The MFS transporter protein having the amino acid sequence of SEQ ID NO: 6 is identified herein as “Fred protein” or “Fred transporter” or “Fred”, interchangeably; a nucleic acid sequence encoding fred protein is identified herein as “Fred coding nucleic acid/DNA” or “fred gene” or “fred”; The amino acid sequence identified herein as SEQ ID NO: 6 is the amino acid sequence that is 100% identical to the amino acid sequence having the GenBank accession ID WP_087817556.1.
(308) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is Fred.
(309) Vag
(310) The MFS transporter protein having the amino acid sequence of SEQ ID NO: 7 is identified herein as “Vag protein” or “Vag transporter” or “Vag”, interchangeably; a nucleic acid sequence encoding vag protein is identified herein as “Vag coding nucleic acid/DNA” or “vag gene” or “vag”; The amino acid sequence identified herein as SEQ ID NO: 7 is the amino acid sequence that is 100% identical to the amino acid sequence having the GenBank accession ID WP_048785139.1.
(311) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is Vag.
(312) Marc
(313) The MFS transporter protein having the amino acid sequence of SEQ ID NO: 8 is identified herein as “Marc protein” or “Marc transporter” or “Marc”, interchangeably; a nucleic acid sequence encoding marc protein is identified herein as “Marc coding nucleic acid/DNA” or “marc gene” or “Marc”; The amino acid sequence identified herein as SEQ ID NO: 8 is the amino acid sequence that is 100% identical to the amino acid sequence having the GenBank accession ID WP_060448169.1.
(314) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein is Marc.
(315) In one or more exemplary embodiments, the sugar efflux transporter and/or MFS transport protein selected from the group consisting of Bad, Nec, YberC, Fred, Vag, and Marc, may be a functional homologue.

(316) In one or more exemplary embodiments, a sugar efflux transporter functional homologue having an amino acid sequence which is at least 70% identical, such as at least 80% identical, such as at least 85% identical, such as at least 90% identical, such as at least 95% identical or such as at least 99% identical to any one of SEQ ID NOs: 3, 4, 5, 6, 7 or 8.
(317) Glycosyltransferases
(318) To be able to synthesize one or more HMOs, the recombinant cell of the described herein comprises at least one recombinant nucleic acid which encodes a functional enzyme with glycosyltransferase activity. The galactosyltransferase gene may be integrated into the genome (by chromosomal integration) of the genetically engineered cell, or alternatively, it may be comprised in a construct that may be integrated into the genome of the genetically engineered cell or inserted into a plasmid DNA and expressed as plasmid borne. If two or more glycosyltransferases are needed for the production of an HMO, e.g. LNT or LNnT, two or more recombinant nucleic acids encoding different enzymes with glycosyltransferase activity may be integrated in the genome, included in a construct and/or expressed from a plasmid, e.g. a β-1,3-N-acetylglucosaminyltransferase (a first recombinant nucleic acid encoding a first glycosyltransferase) in combination with a β-1,3-galactosyltransferase (a second recombinant nucleic acid encoding a second glycosyltransferase) for the production of LNT, where the first and second recombinant nucleic acid can independently from each other be integrated chromosomally or on a plasmid.
(319) A protein/enzyme with glycosyltransferase activity (glycosyltransferase) may be selected in different embodiments from enzymes having the activity of α-1,2-fucosyltransferase, α-1,3-fucosyltransferase, α-1,3/4-fucosyltransferase, α-1,4-fucosyltransferase α-2,3-sialyltransferase, α-2,6-sialyltransferase, β-1,3-N-acetylglucosaminyltransferase, β-1,6-N-acetylglucosaminyltransferase, β-1,3-galactosyltransferase and β-1,4-galactosyltransferase.
(320) For example, the production of 2′-FL requires that the engineered cell expresses an active α-1,2-fucosyltransferase enzyme; for the production of LNT the engineered cell need to express at least two glycosyltransferases, a β-1,3-N-acetylglucosaminyltransferase and a β-1,3-galactosyltransferase; the production of LNFP-I requires that the engineered cell expresses at least one active α-1,2-fucosyltransferase enzyme in combination with at least two glycosyltransferases, a β-1,3-N-acetylglucosaminyltransferase and a β-1,3-galactosyltransferase. Some non-limiting embodiments of proteins having glycosyltransferase activity, which can be encoded by the recombinant genes comprised by the production cell, can be selected from non-limiting examples of Table 1.
(321) Beta-1,3-N-Acetyl-Glucosaminyltransferase
(322) A β-1,3-N-acetyl-glucosaminyltransferase (also known as UDP-GlcNAc:Gala/β-R β-3-N-acetylglucosaminyltransferase) is any protein which comprises the ability of transferring the N-acetyl-glucosamine of UDP-N-acetyl-glucosamine to lactose or another acceptor molecule, in a beta-1,3-linkage. Preferably, a β-1,3-N-acetyl-glucosaminyltransferase used herein does not originate in the species of the genetically engineered cell i.e. the gene encoding the β-1,3-galactosyltransferase is of heterologous origin. Non-limiting examples of β-1,3-N-acetyl-glucosaminyltransferase are given in table 1. β-1,3-N-acetyl-glucosaminyltransferase variants may also be useful, preferably such variants are at least 80%, such as at least 85%, such as at least 90, such as at least 95% identical to one of the β-1,3-N-acetyl-glucosaminyltransferase in table 1.
(323) Production of neutral N-acetylglucosamine-containing HMOs in modified bacteria is also known in the art (see e.g. Gebus C et al. (2012) Carbohydrate Research 363 83-90).
(324) For the production of N-acetylglucosamine-containing HMOs, such as Lacto-N-triose II (LNT-II), Lacto-N-tetraose (LNT), Lacto-N-neotetraose (LNnT), Lacto-N-fucopentaose I (LNFP-I), Lacto-N-fucopentaose II (LNFP-II), Lacto-N-fucopentaose III (LNFP-III), Lacto-N-fucopentaose V (LNFP-V), Lacto-N-fucohexaose V (LNFP-VI), Lacto-N-difucohexaose I (LDFH-I), Lacto-N-difucohexaose II (LDFH-II), and Lacto-N-neodifucohexaose II (LNDFH-III), the

genetically engineered cell comprises a dysfunctional lacZ gene, and it is modified to comprise an exogenous UDP-GlcNAc:Gala/β-R β-3-N-acetylglucosaminyltransferase gene, or a functional homologue or fragment thereof.
(325) This exogenous gene may be obtained from any one of a number of sources, e.g., the IgtA gene described from *N. meningitidis* (Genbank protein Accession AAF42258.1 or WP_033911473.1) or *N. gonorrhoeae* (Genbank protein Accession ACF31229.1).
(326) In an exemplary embodiment of the present invention the genetically engineered cell comprises at least one copy of a β-1,3-N-acetyl-glucosaminyltransferase of SEQ ID NO: 40 or a functional homologue thereof having an amino acid sequence which is at least 80% identical, such as at least 85%, such as at least 90%, such as at least 95% identical to SEQ ID NO: 40. It may be advantageous to have at least to copies of β-1,3-N-acetyl-glucosaminyltransferase in the genetically engineered cell of the present invention.
(327) Optionally, an additional exogenous glycosyltransferase gene may be co-expressed in the bacterium comprising an exogenous UDP-GlcNAc:Gala/β-R β-3-N-acetylglucosaminyltransferase. For example, a β-1,3-galactosyltransferase gene is co-expressed with the UDP-GlcNAc:Gala/β-R β-3-N-acetylglucosaminyltransferase gene to generate an LNT producing cell.
(328) β-1,3-Galactosyltransferase
(329) A β-1,3-Galactosyltransferase is any protein that comprises the ability of transferring the galactose of UDP-Galactose to a N-acetyl-glucosaminyl moiety to an acceptor molecule in a beta-1,3-linkage. Preferably, a β-1,3-galactosyltransferase used herein does not originate in the species of the genetically engineered cell i.e., the gene encoding the β-1,3-galactosyltransferase is of heterologous or exogenous origin. Non-limiting examples of β-1,3-galactosyltransferases are given in table 2. β-1,3-galactosyltransferases variants may also be useful, preferably such variants are at least 80%, such as at least 85%, such as at least 90, such as at least 95% identical to one of the β-1,3-galactosyltransferases in table 1.
(330) This exogenous β-1,3-galactosyltransferase gene can be obtained from any one of a number of sources, e.g., the one described from *H. pylori*, the galTK gene (homologous to Genbank protein Accession BD182026.1), or from the wbgO gene (Genbank protein Accession WP_000582563.1), or from *H. pylori*, the jhp0563 gene (Genbank protein Accession AEZ55696.1), or from *Streptococcus agalactiae* type lb O12 the cpslBJ gene (Genbank protein Accession AB050723. Functional variants and fragments of any of the enzymes described above are also encompassed by the disclosed invention
(331) In an exemplary embodiment of the present invention the genetically engineered cell comprises at least one copy of a β-1,3-galactosyltransferase of SEQ ID NO: 41 or a functional homologue thereof having an amino acid sequence which is at least 80% identical, such as at least 85%, such as at least 90%, such as at least 95% identical to SEQ ID NO: 41. It may be advantageous to have at least to copies of β-1,3-N-acetyl-glucosaminyltransferase in the genetically engineered cell of the present invention.
(332) In an exemplified embodiment, both the first and second recombinant nucleic acids are stably integrated into the chromosome of the production cell; in another presently exemplified embodiment at least one of the first and second glycosyltransferase is plasmid-borne.
(333) Lactose Permease
(334) In the present invention, lactose is used as the substrate for the synthesis of LNT-II, which is then sued as substrate for the synthesis of LNT (or LNnT) which is then used as substrate for the synthesis of LNFP-I. Thus, a genetically engineered cell of the present invention should be capable of importing lactose into the cell. While lactose is naturally imported into some microorganisms, other microorganisms lack the ability to do so. To enable lactose import, such microorganisms would need to be genetically engineered to take up lactose. Thus, in embodiments of the present invention, the genetically engineered cell of the present invention, is able to import lactose into the cell.

(335) One way to enable lactose import into a cell of the present invention is by expression of a lactose permease. In microorganisms comprising a lactose import pathway, the overexpression of an endogenous lactose import pathway, such as but not limited to an endogenous lactose permease protein, and/or incorporation of a heterologous lactose import pathway, such as but not limited to a heterologous lactose permease, may be used to enhance the lactose import of said microorganism. Thus, in embodiments of the present invention, the genetically engineered cell of the present invention overexpresses an endogenous lactose permease protein and/or expresses a heterologous lactose permease.

(336) β-Galactosidase

(337) A genetically engineered cell capable of producing one or more HMOs, e.g. *E. coli*, may comprise an endogenous β-galactosidase gene or an exogenous β-galactosidase gene, e.g. *E. coli* comprises an endogenous lacZ gene (e.g., GenBank Accession ID V00296 (GI:41901)).

(338) An HMO-producing host cell is genetically manipulated to either not comprise any β-galactosidase gene or to comprise the gene that is inactivated. The gene may be inactivated by a complete or partial deletion of the corresponding nucleic acid sequence from the bacterial genome, or the gene sequence is mutated in the way that it is not transcribed, or, if transcribed, the transcript is not translated or if translated to a protein (i.e. β-galactosidase), the protein does not have the corresponding enzymatic activity. In this way the HMO-producing bacterium accumulates an increased intracellular lactose pool which is beneficial for the production of HMOs.

(339) Glycosyl-Donor-Nucleotide-Activated Sugar Pathways

(340) When carrying out the method of this invention, preferably a glycosyltransferase mediated glycosylation reaction takes place in which an activated sugar nucleotide serves as glycosyl-donor. An activated sugar nucleotide generally has a phosphorylated glycosyl residue attached to a nucleoside. A specific glycosyl transferase enzyme accepts only a specific sugar nucleotide. Thus, preferably the following activated sugar nucleotides are involved in the glycosyl transfer: glucose-UDP-GlcNAc, UDP-galactose, UDP-glucose, UDP-N-acetylglucosamine, UDP-N-acetylgalactosamine (GlcNAc) and CMP-N-acetylneuraminic acid. The genetically modified cell according to the present invention can comprise one or more pathways to produce a nucleotide-activated sugar selected from the group consisting of glucose-UDP-GlcNAc, GDP-fucose, UDP-galactose, UDP-glucose, UDP-N-acetylglucosamine, UDP-N-acetylgalactosamine and CMP-N-acetylneuraminic acid (CMP-Neu5Ac). In table 5 below are non-limiting examples of glycosyl-doners and the HMO products they can be used to produce, the list may not be exhaustive.

(341) TABLE-US-00002 TABLE 5 glycosyl-donor HMO product list Glycosyl- donor HMO product UDP-GlcNAc LNT, LNnT, LNFP-I, LNFP-II, LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, LNDFH-II, LNDFH-III, LNH, LNnH, pLNH, pLNnH, F-pLNH-I, F-pLNH-II, F-pLNH-I, F-pLNnH-II, FLSTa, FLSTb, FLSTc, FLSTd, LSTa, LSTb, LSTc, LSTd, DSLNT, SLNH, SLNH-II UDP-Gal LNT, LNnT, LNFP-I, LNFP-II, LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, LNDFH-II, LNDFH-III, LNH, LNnH, pLNH, pLNnH, F-pLNH-I, F-pLNH-II, F-pLNH-I, F-pLNnH-II, FLSTa, FLSTb, FLSTc, FLSTd, LSTa, LSTb, LSTc, LSTd, DSLNT, SLNH-I, SLNH-II GDP-fucose 2′FL, 3FL, DFL, LNFP-I, LNFP-II, LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, LNDFH-II, LNDFH-III, F-LNH, F-LNnH, F-pLNH-I, F-pLNH-II, F-pLNH-I, F-pLNnH-II, FSL, FLSTa, FLSTb, FLSTc, FLSTd CMP-Neu5Ac 3′SL, 6′SL, FSL, FLSTa, FLSTb, FLSTc, FLSTd, LSTa, LSTb, LSTc, LSTd, DSLNT, SLNH-I, SLNH-II

(342) In one embodiment of the method of the invention, the genetically modified cell is capable of producing one or more activated sugar nucleotides mentioned above by a de novo pathway. In this regard, an activated sugar nucleotide is made by the cell under the action of enzymes involved in the de novo biosynthetic pathway of that respective sugar nucleotide in a stepwise reaction sequence starting from a simple carbon source like glycerol, sucrose, fructose or glucose (for a review for monosaccharide metabolism see e.g. H. H. Freeze and A. D. Elbein: Chapter 4: Glycosylation precursors, in: Essentials of Glycobiology, 2nd edition (Eds. A. Varki et al.), Cold

Spring Harbour Laboratory Press (2009)).
(343) The enzymes involved in the de novo biosynthetic pathway of an activated sugar nucleotide can be naturally present in the cell or introduced into the cell by means of gene technology or recombinant DNA techniques, all of them are parts of the general knowledge of the skilled person.
(344) In another embodiment, the genetically modified cell can utilize salvaged monosaccharides for sugar nucleotide. In the salvage pathway, monosaccharides derived from degraded oligosaccharides are phosphorylated by kinases, and converted to nucleotide sugars by pyrophosphorylases. The enzymes involved in the procedure can be heterologous ones, or native ones of the host cell.
(345) Colanic Acid Gene Cluster
(346) For the production of fucosylated HMO's the colanic acid gene cluster is important to ensure presence of sufficient GDP-fucose. In *Escherichia coli* GDP-fucose is an intermediate in the production of the extracellular polysaccharide colanic acid, a major oligosaccharide of the bacterial cell wall. In the context of the present invention the colanic acid gene cluster encodes the enzymes involved in the de novo synthesis of GDP-fucose (gmd, wcaG, wcaH, wcaI, manB, manC), whereas one or several of the genes downstream of GDP-L-fucose, such as wcaJ, can be deleted to prevent conversion of GDP-fucose to colanic acid.
(347) The colanic acid gene cluster responsible for the formation of GDP-fucose comprises or consists of the genes: gmd which encodes the protein GDP-mannose-4,6-dehydratase (UniProt accession nr P0AC88); wcaG (fcl) which encodes the protein GDP-L-fucose synthase (EC 1.1.1.271, UniProt accession nr P32055); wcaH which encodes the protein GDP-mannose mannosyl hydrolase hydrolase (EC 3.6.1.-, UniProt accession nr P32056); wcaI which encodes the colanic acid biosynthesis glycosyltransferase (UniProt accession nr P32057); manB which encodes the protein phosphomannomutase (EC 5.4.2.8, UniProt accession nr P24175) and manC which encodes the protein mannose-1-phosphate guanylyltransferase guanylyltransferase (EC: 2.7.7.13, UniProt accession nr P24174).
(348) In one or more exemplary embodiment(s), the colanic acid gene cluster responsible for the formation of GDP-fucose may be expressed from its native genomic locus. The expression may be actively modulated to increase GDP-fucose formation. The expression can be modulated by swapping the native promoter with a promoter of interest, and/or increasing the copy number of the colanic acid genes coding said protein(s) by expressing the gene cluster from another genomic locus than the native, or episomally expressing the colanic acid gene cluster or specific genes thereof.
(349) In relation to the present disclosure, the term “native genomic locus”, in relation to the colanic acid gene cluster, relates to the original and natural position of the gene cluster in the genome of the genetically engineered cell.
(350) Use of a Genetically Engineered Cell
(351) The disclosure also relates to any commercial use of the genetically engineered cell or the nucleic acid construct, comprising a recombinant nucleic acid sequence which is at least 70% identical to SEQ ID NO: 2, wherein the nucleic acid sequence encodes an α-1,2-fucosyltransferase of SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 80% identical, such as at least 85% identical, or such as at least 90% identical, or such as at least 95% identical to SEQ ID NO: 1.
(352) Thus, in one or more exemplary embodiments, the genetically engineered cell or the nucleic acid construct according to the invention is used in the manufacturing of one or more HMOs, in particular one or more fucosylated HMOs. The one or more HMOs can be selected from the group consisting of 2′-FL, LNT-II, LNT, LNnT, LNFP-I, DFL pLNnH, and LNDFH-I, and the one or more fucosylated HMOs may be selected from the group consisting of 2′-FL, LNFP-I, DFL pLNnH, and LNDFH-I. The one or more fucosylated HMOs may further comprise other HMOs such as LNT-II, LNT, LNnT or even sialylated HMOs.

(353) In a presently preferred embodiment, the one or more HMOs is/are selected from the group consisting of 2′-FL, LNT-II, LNT, LNFP-I and LNDFH-I.
(354) In one or more exemplary embodiments, the genetically engineered cell and/or the nucleic acid construct is used in the manufacturing of more than one HMO(s), wherein the one or more HMOs is/are selected from the group consisting of 2′-FL, LNT, LNFP-I and LNDFH-I.
(355) In another exemplified embodiment, the genetically engineered cell and/or the nucleic acid construct according to the invention, is used in the manufacturing of more than one fucosylated HMO(s), wherein the HMOs are 2′-FL, LNFP-I and LNDFH-I.
(356) In another exemplified embodiment, the genetically engineered cell and/or the nucleic acid construct according to the present disclosure, is used in the manufacturing of more than one HMO(s), wherein the HMOs are 2′-FL, LNT and LNFP-I.
(357) In another exemplified embodiment, the genetically engineered cell and/or the nucleic acid construct according to the invention, is used in the manufacturing of more than one HMO(s), wherein the HMOs are 2′-FL and LNFP-I.
(358) In another exemplified embodiment, the genetically engineered cell and/or the nucleic acid construct according to the invention, is used in the manufacturing of more than one HMO(s), wherein the HMOs are LNT and LNFP-I.
(359) In one or more exemplary embodiments, the genetically engineered cell and/or the nucleic acid construct is used in the manufacturing of LNFP-I.
(360) In one or more exemplary embodiments, the one or more HMOs described above, contains LNFP-I as the predominant HMO. Specifically, LNFP-I constitute more than 70 molar % of the total HMO, such as more than 75%, such as more than 80%, such as more than 85%, such as more than 90%, such as more than 95 molar % of the total HMO.
(361) Manufacturing of HMOs
(362) To produce one or more HMOs, the genetically engineered cell as described herein are cultivated according to the procedures known in the art in the presence of a suitable carbon source, e.g. glucose, glycerol, sucrose, etc., and the produced HMO is harvested from the cultivation media and the microbial biomass formed during the cultivation process. Thereafter, the HMOs are purified according to the procedures known in the art, e.g. such as described in WO2015188834, WO2017182965 or WO2017152918, and the purified HMOs are used as food ingredients, nutraceuticals, pharmaceuticals, or for any other purpose, e.g. for research.
(363) Manufacturing of HMOs is typically accomplished by performing cultivation in larger volumes. The term “manufacturing” and “manufacturing scale” in the meaning of the invention defines a fermentation with a minimum volume of 5 L culture broth. Manufacturing scale or large-scale production are typically a fermentation with a minimum volume of 1,000 L, such as 10,000 L, such as 50,000 L, such as 100,000 L, such as 200,000 L, such as 300,000 L culture broth. Usually, a “manufacturing scale” process is defined by being capable of processing large volumes of a preparation containing the product of interest and yielding amounts of the HMO of interest that meet, e.g. in the case of a therapeutic compound or composition, the demands for clinical trials as well as for market supply. In addition to the large volume, a manufacturing scale method, as opposed to simple lab scale methods like shake flask cultivation, is characterized by the use of the technical system of a bioreactor (fermenter) which is equipped with devices for agitation, aeration, nutrient feeding, monitoring and control of process parameters (pH, temperature, dissolved oxygen tension, back pressure, etc.). To a large extent, the behaviour of an expression system in a lab scale method, such as shake flasks, benchtop bioreactors or the deep well format described in the examples of the disclosure, does allow to predict the behaviour of that system in the complex environment of a manufacturing bioreactor.
(364) With regard to the suitable cell medium used in the fermentation process, there are no limitations. The culture medium may be semi-defined, i.e. containing complex media compounds (e.g. yeast extract, soy peptone, casamino acids, etc.), or it may be chemically defined, without any

complex compounds. Where sucrose is used as the carbon and energy source, a minimal medium might be preferable.
(365) Manufactured Product
(366) The term “manufactured product” according to the use of the genetically engineered cell or the nucleic acid construct refer to the one or more HMOs intended as the one or more product HMO. The various products are described above.
(367) Advantageously, the methods disclosed herein provides both a decreased ratio of by-product to product and an increased overall yield of the product (and/or HMOs in total). This, less by-product formation in relation to product formation facilitates an elevated product production and increases efficiency of both the production and product recovery process, providing superior manufacturing procedure of HMOs.
(368) The manufactured product may be a powder, a composition, a suspension, or a gel comprising one or more HMOs.
(369) TABLE-US-00003 TABLE 1 Non-limiting examples of glycosyltransferases in the framework of the present disclosure Protein Sequence ID Gene (GenBank) Description HMO example IgtA_Nm WP_033911473.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Nm_MC58 AAF42258.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Hd AAN05638.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Ng_PID2 AAK70338.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Ng_NCCP11945 ACF31229.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Past AAK02595.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Nc EEZ72046.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH IgtA_Nm_87255 ELK60643.1 β-1,3-N- LNT-II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH HD0466 WP_010944479.1 β-1,3-N- LNT II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH PmnagT WP_014390683.1 β-1,3-N- LNT II, LNT, LNnT, LNFP-I, LNFP-II, acetylglucosaminyl- LNFP-III, LNFP-V, LNFP-VI, LNDFH-I, transferase LNDFH-II, LNDFH-III, pLNH, F-pLNH I, pLNnH galT_Hp WP_001262061.1 β-1,4-galactosyl- LNnT, LNFP-III, LNFP-VI, LNDFH-III, transferase pLNH I, F-pLNH I, pLNnH wbgO WP_000582563.1 β-1,3-galactosyl- LNT, LNFP-I, LNFP-II, LNFP-V, LNDFH- transferase I, LNDFH-II, pLNH, F-pLNH I cpslBJ AB050723.1 β-1,3-galactosyl- LNT, LNFP-I, LNFP-II, LNFP-V, LNDFH- transferase I, LNDFH-II, pLNH, F-pLNH I jhp0563 AEZ55696.1 β-1,3-galactosyl- LNT, LNFP-I, LNFP-II, LNFP-V, LNDFH- transferase I, LNDFH-II, pLNH, F-pLNH I galTK homologous to β-1,3-galactosyl- LNT, LNFP-I, LNFP-II, LNFP-V, LNDFH- BD182026.1 transferase I, LNDFH-II, pLNH, F-pLNH I WP_111735921.1 Cvb3galT WP_080969100.1 β-1,3-galactosyl- LNT, LNFP-I, LNFP-II, LNFP-V, transferase LNDFH-I, LNDFH-II, pLNH, F-pLNH I futC WP_080473865.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase FucT2_HpUA802 AAC99764.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase FucT2_EcO126t ABE98421.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase FucT2_Hm12198 CBG40460.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase FucT2_Pm9515 ABM71599.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase

FucT2_HpF57 BAJ59215.1 α-1,2-fucosyl- 2'-FL, DFL, LNFP-I, LNDFH-I, DFL transferase FucT54 ADE13114.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase Smob WP_126455392.1 α-1,2-fucosyl- 2′-FL, DFL, LNFP-I, LNDFH-I, DFL transferase Mtun WP_031437198.1 α-1,2-fucosyl- 2′-FL, LNFP-I, LNDFH-I, DFL transferase

(370) TABLE-US-00004 TABLE 2 Genotypes of the strains MP1, MP3 and MP2 Heterologous Protein Sequence Strain ID Genotype ID (GenBank) MP1 MDO, x1 GlcNAcT*, x1 WP_033911473.1 GalTK**, x1 CA***, x1 BD182026.1 (modified).sup.5 futC.sup.4 WP_080473865.1 MP2 MDO, x1 GlcNAcT*, x1 WP_033911473.1 GalTK**, x1 CA***, x1 BD182026.1 (modified).sup.5 fucT54.sup.6 ADE13114.1 MP3 MDO, x1 GlcNAcT*, x1 WP_033911473.1 GalTK**, x1 CA***, x1 BD182026.1 (modified).sup.5 smob.sup.7 WP_126455392.1 *GlcNAcT: IgtA β-1,3-N-acetyloglucosamine transferase (SEQ ID NO: 40) under control of PglpF promoter **GalTK: β-1,3-galactosyltransferase (SEQ ID NO: 41) under control of PglpF promoter ***CA: extra colanic acid gene cluster (gmd-wcaG-wcaH-wcal-manC-manB) at a locus that is different than the native locus (see for example PCT/EP2021/086932 SEQ ID NO: 30) .sup.4futC: α-1,2-fucosyltransferase with NCBI accession No: WP_080473865.1 under control of PglpF promoter .sup.5BD182026.1 (modified): compared to BD182026.1, the applied β-1,3-galactosyltransferase sequence has two deletions of 12 and 30 amino acids and shares 90% identity in the homologous regions .sup.6fucT54: α-1,2-fucosyltransferase with NCBI accession No: ADE13114.1 under control of PglpF promoter .sup.7smob: gene coding for α-1,2-fucosyltransferase of SEQ ID NO: 1 under control of PglpF promoter

(371) TABLE-US-00005 TABLE 3 Genotypes of the strains MP4, MP5, MP6 and MP7 Strain Heterologous Protein Sequence ID Genotype ID (GenBank) MP4 MDO, x2 GlcNAcT, x1 WP_033911473.1 GalTK, x1 CA***, x1 BD182026.1 (modified).sup.5 smob.sup.4 WP_126455392.1 MP5 MDO, x2 GlcNAcT, x1 WP_033911473.1 GalTK, x1 CA***, x1 BD182026.1 (modified).sup.5 smob.sup.4, x1 WP_126455392.1 PglpF-nec.sup.6 WP_092672081.1 MP6 MDO, x2 GlcNAcT, x1 WP_033911473.1 GalTK, x1 CA***, x1 BD182026.1 (modified).sup.5 smob.sup.4, x1 WP_126455392.1 Plac-yberC.sup.7 EEQ08298.1 MP7 MDO, x2 GlcNAcT, x1 WP_033911473.1 GalTK, x1 CA***, x1 BD182026.1 (modified).sup.5 smob.sup.4, x1 WP_126455392.1 Plac-nec.sup.6 WP_092672081.1 *GlcNAcT: Igta β-1,3-N-acetyloglucosamine transferase (SEQ ID NO: 40) **GalTK: β-1,3-galactosyltransferase (SEQ ID NO: 41) ***CA: colanic acid gene cluster (gmd-wcaG-wcaH-wcal-manC-manB) at a locus that is different than the native locus (see for example PCT/EP2021/086932 SEQ ID NO: 30) .sup.4smob: gene coding for α-1,2-fucosyltransferase of SEQ ID NO: 1 .sup.5BD182026.1 (modified): compared to BD182026.1, the applied β-1,3-galactosyltransferase sequence has two deletions of 12 and 30 amino acids and shares 90% identity in the homologous regions .sup.7nec: MFS transporter with GenBank accession ID WP_092672081.1 under control of PglpF promoter .sup.6YberC: MFS transporter with GenBank accession ID EEQ08298.1

(372) TABLE-US-00006 TABLE 6 Genotypes of the strains MP5, MP8 and MP9 Genotype (all genese are Heterologous Protein Strain ID genetically integrated) Sequence ID (GenBank) MP5 MDO, x2 GlcNAcT.sup.1, x1 GalTK.sup.2, WP_033911473.1 x1 CA.sup.3, x1 smob.sup.4, BD182026.1 (modified).sup.5 x1 PglpF-nec.sup.6 WP_126455392.1 WP_092672081.1 MP8 MDO, x2 GlcNAcT.sup.1, x1 GalTK.sup.2, WP_033911473.1 x1 CA.sup.3, x1 smob.sup.4, x1 BD182026.1 (modified).sup.5 PglpF-nec.sup.6, x1 PglpF-sacC_Agal.sup.7 WP_126455392.1 WP_092672081.1 WP_103853210.1 MP9 MDO, x2 GlcNAcT.sup.1, x1 GalTK.sup.2, WP_033911473.1 x1 CA.sup.3, x1 smob.sup.4, x1 BD182026.1 (modified).sup.5 PglpF-nec.sup.6, x2 PglpF-sacC_Agal.sup.7 WP_126455392.1 WP_092672081.1 WP_103853210.1 .sup.1GlcNAcT: gene coding for the β-1,3-N-acetyloglucosamine transferase LgtA .sup.2GalTK: gene coding for the β-1,3-galactosyltransferase GalTK .sup.3CA: extra colanic acid gene cluster (gmd-wcaG-wcaH-wcal-manC-manB) at a locus that is different than the native locus (see for

example PCT/EP2021/086932 SEQ ID NO: 30) .sup.4smob: gene coding for α-1,2-fucosyltransferase of SEQ ID NO: 1 .sup.5BD182026.1 (modified): compared to BD182026.1, the applied β-1,3-galactosyltransferase sequence has two deletions of 12 and 30 amino acids and shares 90% identity in the homologous regions .sup.6nec: MFS transporter with GenBank accession ID GenBank accession ID WP_092672081.1 under control of PglpF promoter .sup.7sacC_Agal: sucrose invertase with GenBank accession ID GenBank accession WP_103853210.1 (SEQ ID NO: 13)

General

(373) It should be understood that any feature and/or aspect discussed above in connections with the described methods apply by analogy to the cells, nucleic acid constructs and uses thereof described herein.

(374) The terms culturing and fermentation are used interchangeably.

(375) The terms Lacto-N-triose, LNT-II, LNT II, LNT2 and LNT 2, are used interchangeably.

(376) The terms genetically modified and genetically engineered are used interchangeably.

(377) Each specific variation of the features disclosed herein can be applied to all other embodiments of the disclosure unless specifically stated otherwise.

(378) Generally, all terms used herein are to be interpreted according to their ordinary meaning in the technical field, and applicable to all aspects and embodiments of the disclosure, unless explicitly defined or stated otherwise.

(379) All references to "a/an/the [cell, sequence, gene, transporter, step, etc]" are to be interpreted openly as referring to at least one instance of said cell, sequence, gene, transporter, step, etc., unless explicitly stated otherwise.

(380) The following figures and examples are provided below to illustrate the present disclosure. They are intended to be illustrative and are not to be construed as limiting in any way.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. **1** The choice of the α-1,2-fucosyltransferase that is introduced in an *E. coli* DH1 K12 strain producing LNT has a large impact on the HMO content of the final HMO blend. % change in the final LNFP-I titer of fucT54- and smob-expressing cells relative to cells expressing futC.

(2) FIG. **2** LNFP-I to 2′-FL molar ratios for strains expressing the FutC, FucT54 or Smob enzymes.

(3) FIG. **3** Molar % fraction of each HMO in the final HMO blend acquired by strain MP1 expressing the futC gene.

(4) FIG. **4** Molar % fraction of each HMO in the final HMO blend acquired by strain MP2 expressing the fucT54 gene.

(5) FIG. **5** Molar % fraction of each HMO in the final HMO blend acquired by strain MP3 expressing the smob gene.

(6) FIG. **6** The final HMO titers for smob-expressing strains that bear a genomic copy of the nec (strains MP5 and MP7) or yberC (strain MP6) genes, shown relative to the final HMO titers of smob-expressing cells that do not express an MFS transporter (strain MP4), as revealed by the analysis of total samples. The reference level (given as 100%) is shown for strain MP4.

(7) FIG. **7** Fraction of LNFP-I (in molar %) in the final HMO blend for smob-expressing cells that do not express a MFS transporter (strain MP4) and cells that bear a genomic copy of the nec (strains MP5 and MP7) or yberC (strain MP6) genes, as revealed by the analysis of total samples.

(8) FIG. **8** The final HMO titers for smob-expressing strains that bear a genomic copy of the nec (strains MP5 and MP7) or yberC (strain MP6) genes, shown relative to the final HMO titers of smob-expressing cells that do not express an MFS transporter (strain MP4), as revealed by the analysis of the supernatant fraction of the corresponding cultures. The reference level (given as

100%) is shown for strain MP4.
(9) FIG. **9** Fraction of LNFP-I detected in the supernatant (in % of total LNFP-I) in cultures of smob-expressing cells that do not express a MFS transporter (strain MP4) and smob-expressing cells that bear a genomic copy of the nec (strains MP5 and MP7) or yberC (strain MP6) genes.
(10) FIG. **10** Pathways for producing LNFP-I and 2′-FL respectively from lactose. 2′FL is produced in a single step from lactose in the presence of the enzyme α-1,2-fucosyltransferase (α-1,2-ft) adding fucose to the lactose. Production of LNFP-I is a 3 step process where a β-1,3-N-acetyl-glucosaminyltransferase (β-1,3-GlcNacT) adds N-acetylglucosamine to lactose to form LNT-II to which a β-1,3-galactosyltransferase (β-1,3-GalT) adds galactose forming LNT on which an α-1,2-fucosyltransferase (α-1,2-ft) adds a fucose to form LNFP-I. As illustrated in example 1 different α-1,2-fucosyltransferase may have different substrate specificities, i.e. FutC seem to have higher specificity for lactose whereas smob seems to have higher specificity for LNT as substrate.
(11) FIG. **11** Batch growth profile on sucrose for *E. coli* cells that do or do not express protein(s) that enable the utilization of sucrose as the main and/or the sole, carbon source and/or energy source. The strain MP5 is not capable of utilizing sucrose, while the strains MP8 and MP9 express the extracellular sucrose hydrolase SacC_Agal from one or two PglpF-driven genomic copies, respectively

EXAMPLES

Example 1—The LNFP-I Content in Neutral HMO Blends can be Modulated by the Choice of the Expressed α-1,2-Fucosyltransferase

(12) Description of the Genotype of Strains MP1, MP2 and MP3 Tested in Deep Well Assays
(13) The strains (genetically engineered cells) constructed in the present application were based on *Escherichia coli* K-12 DH1 with the genotype: F.sup.−, λ.sup.−, gyrA96, recA1, relA1, endA1, thi-1, hsdR17, supE44. Additional modifications were made to the *E. coli* K-12 DH1 strain to generate the platform strain “MDO” with the following modifications: lacZ: deletion of 1.5 kbp, lacA: deletion of 0.5 kbp, nanKETA: deletion of 3.3 kbp, melA: deletion of 0.9 kbp, wcaJ: deletion of 0.5 kbp, mdoH: deletion of 0.5 kbp, and insertion of Plac promoter upstream of the gmd gene.
(14) Methods of inserting or deleting gene(s) of interest into the genome of *E. coli* are well known to the person skilled in the art. Insertion of genetic cassettes into the *E. coli* chromosome can be done using gene gorging (see e.g., Herring and Blattner 2004 J. Bacteriol. 186:2673-81 and Warming et al 2005 Nucleic Acids Res. 33(4):e36) with specific selection marker genes and screening methods.
(15) Based on the platform strain (“MDO”), the modifications summarised in Table 2, were made to obtain the fully chromosomal strains MP1, MP3 and MP2. The strains can produce the pentasaccharide HMO, LNFP-I. The glycosyltransferase enzymes LgtA (a β-1,3-N-acetyloglucosamine transferase) from *N. meningitidis* and GalTK (a β-1,3-galactosyltransferase) from *H. pylori* are present in all three strains.
(16) In addition, each of these strains bears a single genomic copy of a gene encoding an α-1,2-fucosyltransferase as indicated in table 2, whose expression is driven by the synthetic inducible promoter PglpF.
(17) Specifically, the strain MP1 expresses a single PglpF-driven copy of the futC gene (*Helicobacter pylori* 26695, Wang et al, Mol. Microbiol., 1999, 31, 1265-1274, GenBank ID: WP_080473865.1), the strain MP3 expresses a single PglpF-driven copy of the fucT54 gene (Sideroxydans lithotrophicus ES-11, WO2019008133A1, GenBank ID: WP_013031010.1) and the strain MP3 expresses a single PglpF-driven copy of the smob gene (*Sulfuriflexus mobilis*, SEQ ID NO: 1 of the present disclosure).
(18) The present Example describes for the first time an α-1,2-fucosyltransferase that is found in nature, Smob, and shows an unpreceded high specificity for LNT and simultaneously a very low specificity for lactose. Contrary to previously tested α-1,2-fucosyltransferases, cells that concomitantly express two glycosyltransferases (SEQ ID NO: 40 and 41) required for LNT

synthesis and the Smob enzyme produce almost exclusively LNFP-I. In this manner, the present disclosure demonstrates how the simple strain engineering approach of introducing a single heterologous gene, smob, into the genome of an *E. coli* DH1 K12 strain that already produces LNT can be advantageously employed to either increase the LNFP-I content of neutral HMO blends or establish an in vivo production process that results in an almost “pure” LNFP-I HMO product.
(19) See: Table 2. Genotypes of the strains MP1, MP3 and MP2
(20) Description of the Applied Deep Well Assay Protocol for Strain Characterization
(21) The strains disclosed in the present example were screened in 96 deep well plates using a 4-day protocol. During the first 24 hours, precultures were grown to high densities and subsequently transferred to a medium that allowed induction of gene expression and product formation. More specifically, during day 1, fresh precultures were prepared using a basal minimal medium supplemented with magnesium sulphate, thiamine and glucose. The precultures were incubated for 24 hours at 34° C. and 1000 rpm shaking and then further transferred to a new basal minimal medium (BMM, pH 7.5) in order to start the main culture. The new BMM was supplemented with magnesium sulphate, thiamine, a bolus of 20% glucose solution (50 ul per 100 mL) and a bolus of 10% lactose solution (5 ml per 100 ml). Moreover, 50% sucrose solution was provided as carbon source, accompanied by the addition of sucrose hydrolase (invertase), so that glucose was released at a rate suitable for C-limited growth. The main cultures were incubated for 72 hours at 28° C. and 1000 rpm shaking.
(22) For the analysis of total broth, the 96-well plates were boiled at 100° C., subsequently centrifuged, and finally the supernatants were analysed by HPLC. For supernatant samples, the initial centrifugation of microtiter plates was followed by the removal of 0.1 mL supernatant for direct analysis by HPLC. For pellet samples, the cells were initially washed, then dissolved in deionized water and centrifuged. Following centrifugation, the pellets were analysed for HMO content in the cell interior after resuspension, boiling, centrifugation and analysis of the final supernatant.
(23) Results of the Deep Well Assays Since the FutC enzyme is known to have a higher specificity for lactose than for LNT, the formation of 2′-FL is favoured in futC-expressing cells (Wang et al, Mol. Microbiol., 1999, 31, 1265-1274). To promote LNFP-I synthesis in vivo, it is therefore desirable to identify α-1,2-fucosyltransferases other than FutC that show higher specificity for LNT than for lactose. This has been attempted before in patent WO2019008133A1, where several α-1,2-fucosyltransferases were associated with a high specificity for LNT, including the FucT54 enzyme.
(24) In our experiments, strains that express the FutC, FucT54 or Smob enzymes were constructed and characterized in deep well assays, and samples were collected from the total broth of the cultures. All samples were analysed for HMO content by HPLC following the 72-hour protocol described above. The concentration of the detected HMOs in each sample was used to calculate the relative differences in the HMO content of the strains tested, i.e., the % HMO content of fucT54- and smob-expressing cells relative to the HMO content of futC-expressing cells. LNFP-I to 2′-FL ratios were also calculated based on the HMO concentrations determined by HPLC analysis. Finally, the molar % fraction of each HMO in the final blend acquired by each strain was calculated to report the overall HMO profile of each strain.
(25) As revealed by the analysis of the total samples in deep-well cultures and the calculations mentioned above, the enzyme identified in the present disclosure, namely Smob, appears to be superior to the FutC and FucT54 enzymes in terms of producing LNFP-I due to its inherent high specificity for LNT. As shown in FIG. **1**, final LNFP-I titers of the strains expressing the fucT54 gene (strain MP2) and the smob gene (strain MP3) are, respectively, 40% and 70% higher than the LNFP-I titer reached with the strain expressing the futC gene (strain MP1). Likewise, the 1:1 LNFP-1/2′-FL ratio in futC-expressing cells is changed to 2:1 in fucT54- and 11:1 in smob-expressing cells (FIG. **2**). As shown in FIG. **3-5**, the HMO content in the blends acquired by cells expressing the futC, fucT54 or smob genes differs markedly from strain to strain. Specifically, the

almost 50%:50% LNFP-I:2′FL content in the HMO blend delivered by the strain MP1 (FutC) is turned to almost 70%:30% in the strain MP2 (FucT54) or even 90%:10% in the strain MP3 (Smob).
(26) These results highlight the unique advantage offered by the Smob enzyme to generate neutral HMO blends enriched in LNFP-I, or cells producing almost exclusively LNFP-I, which is a highly desired strain engineering goal. Also, the fact that the Smob enzyme meets the above expectations to a higher degree than the previously identified FucT54 enzyme (WO2019008133A1) is unexpected and thereby strengthens the impact of the present disclosure on the HMO field.
Example 2—The Concomitant Expression of the Smob Enzyme and Either of the Heterologous MFS Transporters Nec or YberC is the Key for an Efficient LNFP-I Cell Factory
(27) Description of the Genotype of Strains MP4, MP5, MP6 and MP7 Tested in Deep Well Assays
(28) Based on the platform strain ("MDO") described in example 1, the modifications summarised in Table 3, were made to obtain the fully chromosomal strains MP4, MP5, MP6 and MP7. The strains can produce the pentasaccharide HMO LNFP-I. The glycosyltransferase enzymes LgtA (a β-1,3-N-acetyloglucosamine transferase, SEQ ID NO: 40) from *N. meningitidis*, GalTK (a β-1,3-galactosyltransferase, SEQ ID NO: 41) from *H. pylori* and Smob (α-1,2-fucosyltransferase (SEQ ID NO: 1) from *S. mobilis* are present in all four strains. Moreover, the strain MP6 expresses the heterologous transporter of the Major Facilitator Superfamily (MFS) YberC (SEQ ID NO: 5) from *Yersinia bercovieri*, while the strains MP5 and MP7 express the heterologous MFS transporter Nec (SEQ ID NO: 4) from *Rosenbergiella nectarea*. The only difference between the latter two strains lies in the strength of the promoter that drives the expression of the nec gene, i.e. a PglpF-driven nec copy is present in the strain MP5, while the strain MP7 expresses the nec gene under the control of the Plac promoter.
(29) The present Example describes an optimized strain engineering approach to construct a highly efficient LNFP-I cell factory that produces LNFP-I at high titers, with a significant fraction of the product being found in the supernatant of the culture. Following the approach described here, HMOs other than LNFP-I constitute only a minor fraction of the total HMO blend delivered by the engineered cell. In the framework of the present Example, introducing the heterologous genes, smob and nec or yberC, ito the genome of an *E. coli* DH1 K12 strain that already produces LNT can be advantageously employed with a high copy number for the IgtA gene to deliver an efficient LNFP-I cell factory with the beneficial traits described above.
(30) See: Table 3. Genotypes of the strains MP4, MP5, MP6 and MP7
(31) Description of the Applied Deep Well Assay Protocol for Strain Characterization
(32) The strains disclosed in the present example were screened in 96 deep well plates using a 4-day protocol. During the first 24 hours, precultures were grown to high densities and subsequently transferred to a medium that allowed induction of gene expression and product formation. More specifically, during day 1, fresh precultures were prepared using a basal minimal medium supplemented with magnesium sulphate, thiamine and glucose. The precultures were incubated for 24 hours at 34° C. and 1000 rpm shaking and then further transferred to a new basal minimal medium (BMM, pH 7.5) in order to start the main culture. The new BMM was supplemented with magnesium sulphate, thiamine, a bolus of 20% glucose solution (50 ul per 100 mL) and a bolus of 20% lactose solution (10 ml per 175 ml). Moreover, 50% sucrose solution was provided as carbon source, accompanied by the addition of sucrose hydrolase (invertase), so that glucose was released at a rate suitable for C-limited growth. The main cultures were incubated for 72 hours at 28° C. and 1000 rpm shaking.
(33) For the analysis of total broth, the 96-well plates were boiled at 100° C., subsequently centrifuged, and finally the supernatants were analysed by HPLC. For supernatant samples, the initial centrifugation of microtiter plates was followed by the removal of 0.1 mL supernatant for direct analysis by HPLC. For pellet samples, the cells were initially washed, then dissolved in deionized water and centrifuged. Following centrifugation, the pellets were analysed for HMO

content in the cell interior after resuspension, boiling, centrifugation and analysis of the final supernatant.
(34) Results of the Deep Well Assays
(35) The expression of many host genes as well as the heterologous genes encoding enzymes involved in the in vivo synthesis of a HMO of interest needs to be fine-tuned to achieve an optimal fermentation output, where the desired HMO product is formed at high titers while molecules other than the main product (i.e., precursor or heavily decorated sugars) are formed in minimal amounts. Moreover, the export of the newly formed HMO of interest needs to be exported in the cell exterior to alleviate the cell from the HMO-imposed osmotic stress. The identification of sugar exporters and the fine balancing of their expression can be a key for the success of such production systems. This task can though be challenging, since only the HMO of interest, and not the precursor or elongated versions thereof, should be bound and exported by the chosen sugar exporter.
(36) After following some strain engineering rounds to balance the expression of the enzymes involved in LNFP-I synthesis, sugar transporters that were proven to be able to export the LNFP-I product out of the cell, namely Nec and YberC (FIG. **9**), were also introduced in the LNFP-I production system. Relevant strains were constructed and characterized in deep well assays as described in the previous sections. Samples were collected from the total broth as well as the supernatant and pellet fractions of the cultures. All samples were analysed for HMO content by HPLC following the 72-hour protocol described above. The concentration of the detected HMOs in each sample was used to calculate the % quantitative differences in the HMO content of the strains tested, i.e., the % HMO content of nec- and yberC-expressing cells relative to the HMO content of cells that do not express a heterologous transporter.
(37) As revealed by the analysis of the total samples in deep-well cultures, minor gains in LNFP-I titers can be achieved when a transporter is expressed in a LNFP-I system with balanced expression of glycosyltransferases, i.e. up to 10% higher LNFP-I titers are observed for then strain MP7 (Nec) than the strain MP4 (no transporter) (FIG. **6**). Moreover, the introduction of a sugar exporter in LNFP-I production strains induces drastic changes in the abundance of the other HMOs in the final blend. Specifically, LNT II is absent from the total broth of all transporter-expressing cells, with the latter reaching approximately 20% lower 2′-FL and up to 40% higher LNT titers (strains MP6 and MP5) compared to the strain that does not express a sugar transporter (strain MP4) (FIG. **6**). It should be noted that the relatively large increase in LNT titers (up to 40%) observed for the transporter-expressing cells can be attributed to the low values of LNT concentrations reported for all four strains.
(38) The minor gains in the relative final LNFP-I titers and the drastic changes in the relative abundance of other HMOs observed for MFS-expressing cells was expectedly reflected in the absolute LNFP-I fraction (%) that was measured in the final HMO blend. Specifically, as revealed by the analysis of the total samples, the absolute fraction of LNFP-I in the final HMO blend reached approximately 88% for cells expressing the Nec transporter (strains MP5 and MP7), while this fraction corresponded to approximately 81% for cells that do not express a MFS transporter (FIG. **7**).
(39) The analysis of the supernatant and total samples of strains tested in deep-well cultures revealed similar trends regarding the observed relative abundance changes in LNT II, LNT and 2′-FL for transporter-expressing cells (FIGS. **6** and **8**). However, the analysis of supernatant samples revealed a striking fact about LNFP-I. In detail, although only a minor relative overall gain in LNFP-I titer can be achieved by the strain MP7, which expresses the Nec transporter under the control of the Plac promoter (FIG. **6**), the extracellular LNFP-I fraction of this strain is much higher (up to 30%) than the strain MP4, which does not express a sugar transporter (FIG. **8**). The same trend is observed for the strains expressing the YberC transporter (strain MP6) or the Nec transporter from a PglpF-driven genomic copy (strain MP5) (FIG. **8**).
(40) The marked increase in the supernatant fraction of LNFP-I for MFS-relative to no MFS-

expressing cells was expectedly reflected in the absolute LNFP-I fraction (%) that was detected in the supernatant of the corresponding cultures. In detail, only 24% of the total LNFP-I was detected in the supernatant for cells that do not express an MFS transporter (strain MP4), while approximately 38% of the synthesized LNFP-I was detected in the supernatant of cultures for cells expressing the Nec transporter (FIG. **9**).

(41) In conclusion, the balanced expression of the β-1,3-N-acetylglucosamine transferase LgtA, the β-1,3-galactosyltransferase GalTK, the α-1,2-fucosyltransferase Smob and either of the MFS transporters Nec or YberC constitute an effective strain engineering strategy for the generation of a highly productive LNFP-I cell factories. Such microbial systems produce LNFP-I at high titers, with a significant fraction of the product being found in the supernatant of the culture, and HMOs other than LNFP-I representing only a minor fraction of the total HMO blend delivered by the engineered cell.

Example 3—The SacC_Agal Sucrose Utilization Technology can be Successfully Applied to Engineer *E. coli* Cells Producing the Complex Pentasaccharide LNFP-I

(42) Description of the Genotype of Strains MP5, MP8 and MP9

(43) Based on the platform strain (“MDO”, MP1) described in example 1, the modifications summarised in Table 6 were made to obtain the fully chromosomal strains MP5, MP8 and MP9.

(44) The strains can produce the pentasaccharide HMO LNFP-I, the tetrasaccharide HMO LNT and the trisaccharide HMO 2′-FL. The glycosyltransferase enzymes LgtA (a β-1,3-N-acetylglucosamine transferase) from *N. meningitidis*, GalTK (a β-1,3-galactosyltransferase) from *H. pylori*, Smob (α-1,2-fucosyltransferase) from *S. mobilis* and the heterologous MFS transporter Nec from *Rosenbergiella nectarea*. are present in all three strains. Contrary to the strain MP5, the strains MP8 and MP11 can utilize sucrose as the carbon and energy source since the gene sacC_Agal from *Avibacterium gallinarum* is integrated on their genome in one or two loci, respectively.

(45) This invention demonstrates how the introduction of an extracellular invertase such as SacC_Agal can be advantageously used to confer an engineered *E. coli* that produces the complex pentasaccharide LNFP-I the ability to utilize sucrose as carbon and/or energy source. The only difference between the strains MP5 and MP8 or MP9, as shown in the table below, is the absence of the SacC_Agal enzyme from the former and its presence in the latter two strains. Although the strain MP8 bears a single PglpF-driven copy of the sacC_Agal gene, the strain MP9 bears two such copies.

(46) In the present Example, it is demonstrated that sacC_Agal-expressing cells not only grow robustly in batch cultures containing sucrose, but they also produce LNFP-I at high titers in fed-batch fermentation processes.

(47) Description of the Protocol Applied During Growth Monitoring Assays

(48) The strains disclosed in the present example were screened in 96 well microtiter plates using a 2,5-day protocol. During the first 24 hours, cells were grown to high densities while in the next 36 hours cells were transferred to a medium containing sucrose as the main carbon and energy source. Specifically, during day 1, fresh inoculums were prepared using a Luria-Bertani broth containing 20% glucose. After 24 hours of incubation of the prepared cultures at 34° C., cells were transferred to a basal minimal medium (200 uL) supplemented with magnesium sulphate and thiamine to which an initial bolus of 20% glucose solution and 15 g/L sucrose solution as carbon source was provided to the cells. After inoculation of the new medium, cells were shaken at 1200 rpm at 28° C. for 72 hours. The cells were grown in a batch mode of cultivation in microtiter plates that were compatible with the Varioskan LUX Multimode Microplate Reader from ThermoFisher Scientific.

(49) Fermentation Protocol

(50) The *E. coli* strains were cultivated in 250 mL fermenters (Ambr250 HT Bioreactor system, Sartorius) starting with 100 mL of mineral culture medium consisting of 30 g/L glucose or sucrose (AL-X16 and AL-X17 respectively) and a mineral medium comprised of

$NH_4H_2PO_4$, KH_2PO_4, $MgSO_4\times7H_2O$, NaOH, citric acid, trace element solution, antifoam and thiamine. The dissolved oxygen level was kept at 20% by a cascade of first agitation and then airflow starting at 700 rpm (up to max 4500 rpm) and 1 VVM (up to max 3 VVM). The pH was kept at 6.8 by titration with 8.5% NH_4OH solution. The cultivations were started with 2% (v/v) inoculums from pre-cultures comprised of 10 g/L glucose (AL-X16) or sucrose (AL-X17), $(NH_4)_2HPO_4$, KH_2PO_4, $MgSO_4\times7H_2O$, KOH, NaOH, citric acid, trace element solution, antifoam and thiamine. After depletion of the glucose or sucrose contained in the basal minimal medium, a glucose (AL-X16) or sucrose-(AL-X17) containing feed solution was continuously added to the fermenter at a rate that maintained carbon-limiting conditions. The temperature was initially at 33° C. but was dropped to 30° C. after 3 hours of feeding. Lactose was added as a bolus addition of 25% lactose monohydrate solution 36 hours after feed start and then every 19 hours to keep lactose from being a rate limiting factor. The growth, metabolic activity and metabolic state of the cells was followed by on-line measurements of reflectance and CO_2 evolution rate. Throughout the fermentations, samples were taken to determine the concentration of HMO products, lactose and other minor by-products using HPLC.
(51) Results of the Growth Monitoring in Assays
(52) Strains were tested in growth monitoring assays using the 60-hour protocol described above, with the cultures being operated at the batch mode in the presence of sucrose. To evaluate the ability of different strains to grow on sucrose as a function of their genetic makeup (i.e., expression or not of the SacC_Agal enzyme that is directly associated with sucrose utilization), the raw data on culture absorbance (in 600 nm), reported by the Varioskan LUX system, was used to inspect the growth curves on sucrose for the strains tested, namely MP5, MP8 and MP9. The data analysis software Skanlt was used to extract all growth curves and execute various calculations.
(53) As shown in FIG. **11**, the strain MP5, which does not bear the sacC_Agal on its genome, cannot grow on the sucrose that is provided in the medium being present in the prepared batch cultures. After a little growth (due to the provided low levels of glucose and the potentially partially degraded sucrose that could be present in the medium), the strain MP5 has a flat growth profile (FIG. **11**). On the contrary, the strain MP8 and MP9, which bear a single or two PglpF-driven copies of the sacC_Agal gene, respectively, grow nicely on sucrose over time and reach much higher optical density values than the strain MP5 (FIG. **10**). It is also noteworthy that the strains MP8 and MP9 have an almost identical growth profile, which indicates that a single PglpF-driven copy of the sacC_Agal gene should be sufficient to support robust growth on sucrose (FIG. **11**).
(54) In this manner, the present disclosure indicates an efficient strain engineering tool for producing flexi-fuel strains (capable of growing on more than one carbon source) with a normal cell physiology, which could indicate a presumably low metabolic burden in sacC_Agal-expressing cells compared to other multi-gene sucrose utilization technologies that are known in the art (e.g., scrBRYA).
(55) In the present example the SacC_Agal sucrose invertase was introduced into the LNFP-I expressing host cell, identified in table 4 as MP5. In detail, the sacC_Agal gene was placed under control of the PglpF promoter and integrated in the chromosome in a single (strain MP8) or two copies (strain MP9). Also, it is hereby demonstrated that sacC_Agal-expressing cells not only grow robustly in batch cultures containing sucrose, but they also produce LNFP-I at high titers in fed-batch fermentation processes as shown in the fermentation results below.
(56) Fermentation Results
(57) The production of LNFP-I, LNT, 2′FL and LNT-II is shown as the fraction % of the total HMO produced. A single fermentation was run with the strain MP5, while the fermentations of the strain MP9 were done in duplicate. The fermentation end-point data is presented in Table 5. In general, in the selected fermentation processes, both strains MP5 and MP9 were producing LNFP-I at high levels and similar titers.
(58) Specifically, the strain that cannot grow on sucrose, namely MP5, provided an HMO profile

that consisted of 3 HMOs when a glucose-based process (AL-X16) was implemented. In detail, the HMO profile of the strain MP5 contains approximately 95% LNFP-I, 1% LNT and 4% 2′-FL (Table 5). The strain expressing sacC_Agal under the control of the PglpF promoter (strain MP9) produced a bit higher amount of 2′-FL compared to the strain MP5 when a sucrose-based process (AL-X17) was implemented. In particular, the HMO profile of the strain MP9 contains approximately 90% LNFP-I, and 10% 2′-FL, but no LNT (Table 5).
(59) TABLE-US-00007 TABLE 5 HMO blend composition in total broth sample at fermentation timepoint 89 h. The strain MP5 grows on glucose (process AL-X16) while the strain MP9 (two independent runs) expresses the SacC_Agal enzyme and can thus utilize sucrose as the carbon and/or energy source (process AL-X17). Fermentation LNFP-I/ 2′-FL/ LNT/ LNT-II/ Batch ID Process HMO HMO HMO HMO GDF22xxx Strain ID (%) (%) (%) (%) 240 MP5 AL-X16 94.9 4.2 0.9 0.0 243 MP9 AL-X17 90.5 9.5 0.0 0.0 244 MP9 AL-X17 90.3 9.7 0.0 0.0
(60) In conclusion, the SacC_Agal sucrose utilization technology enables the high-level LNFP-I production using an accordingly engineered cell, which provides an HMO profile that is highly similar to the one obtained using glucose as the carbon source.
Sequences
(61) The current application contains a sequence listing in text format and electronical format which is hereby incorporated by reference as are the sequences listed in the corrected sequence list in the priority application DK PA 2021 70250. Below is a summary of the sequences which are not presented in Table 4. SEQ ID NO: 1 [smob protein] SEQ ID NO: 2 [smob gene] SEQ ID NO: 3 [bad] SEQ ID NO: 4 [nec] SEQ ID NO: 5 [YberC] SEQ ID NO: 6 [Fred] SEQ ID NO: 7 [Vag] SEQ ID NO: 8 [Marc] SEQ ID NO: 9 [scrY] SEQ ID NO: 10 [scrA] SEQ ID NO: 11 [scrB] SEQ ID NO: 12 [scrR] SEQ ID NO: 13 [SacC_Agal protein] SEQ ID NO: 14 [Bff protein] SEQ ID NO: 40 [IgtA] SEQ ID NO: 41 [GalTK] SEQ ID NO: 42 [GlpR]

Claims

1. A genetically engineered cell capable of producing lacto-N-fucopentaose I (LNFP-I) comprising a recombinant nucleic acid sequence encoding an α-1,2-fucosyltransferase protein comprising SEQ ID NO: 1, or a functional homologue thereof having an amino acid sequence which is at least 90% identical to SEQ ID NO: 1, wherein the α-1,2-fucosyl-transferase is capable of transferring of a fucosyl moiety from a donor molecule onto a galactose moiety of lacto-N-tetraose (LNT) through an α-1,2 coupling.
2. The genetically engineered cell according to claim 1, wherein the predominant fucosylated human milk oligosaccharide (HMO) produced by the cell is LNFP-I.
3. The genetically engineered cell according to claim 1, wherein the cell further comprises a nucleic acid sequence encoding a β-1,3-N-acetyl-glucosaminyltransferase protein and a β-1,3-galactosyltransferase protein.
4. The genetically engineered cell according to claim 1, further comprising a recombinant nucleic acid sequence encoding the major facilitator superfamily (MFS) transporter protein Nec comprising SEQ ID NO: 4 or YberC comprising SEQ ID NO: 5, or a functional homologue thereof having an amino acid sequence which is at least 90% identical to any one of SEQ ID NOs: 4 or 5.
5. The genetically engineered cell according to claim 1, wherein the cell is selected from the group consisting of *Escherichia coli*, *Corynebacterium glutamicum*, *Lactococcus lactis*, *Bacillus subtilis*, *Streptomyces lividans*, *Pichia pastoris*, and *Saccaromyces cerevisiae*.
6. The genetically engineered cell according to claim 1, wherein the expression of the recombinant nucleic acid is regulated by a promoter sequence selected from the group consisting of SEQ ID NO: 16-38 and 39.
7. The genetically engineered cell according to claim 6, wherein the promoter sequence is selected from the group consisting of SEQ ID NO: 27, 22, 28, 29, 30, 32, 33, 34, 37, 38, or 39.

8. The genetically engineered cell according to claim 5, wherein the genetically engineered cell is capable of utilizing sucrose as sole carbon and energy source.
9. The genetically engineered cell according to claim 8, wherein the genetically engineered cell utilizes sucrose by expressing a polypeptide capable of hydrolyzing sucrose into fructose and glucose selected from the group consisting of the SEQ ID NOs: 13 or 14, or a functional homologue thereof having an amino acid sequence which is at least 90% identical to any one of SEQ ID NOs: 13 or 14.
10. A nucleic acid construct comprising a recombinant nucleic acid sequence which is at least 85% identical to SEQ ID NO: 2, wherein the expression of the nucleic acid construct is regulated by a promoter sequence selected from the group consisting SEQ ID NO: 16-38 and 39.
11. A method for producing one or more fucosylated human milk oligosaccharides (HMOs), the method comprising the steps of, a. providing a genetically engineered cell of claim 9, b. culturing the cell in a suitable cell culture medium to express; and c. harvesting one or more fucosylated HMOs produced.
12. The method according to claim 11, wherein the one or more fucosylated HMOs are selected from the group consisting of 2′-fucosyllactose (2′-FL), lacto-N-fucopentaose I (LNFP-I), difucosyllactose (DFL) and lacto-N-difucohexaose I (LNDFH-I).
13. The method according to claim 11, wherein the method predominantly produces LNFP-1.
14. The method according to claim 11, wherein the one or more fucosylated HMOs comprise LNFP-I and 2′-FL, and wherein the method produces a molar ratio of LNFP-I: 2′-FL in the harvested HMOs in step c) in the range of 20:1-2:1.
15. The method according to claim 11, wherein the one or more fucosylated HMOs comprise LNFP-I, and wherein the method produces a molar ratio of LNFP-I: LNT in the harvested HMOs in step c), in the range of 1000:1 to 10:1.
16. The method according to claim 11, wherein the genetically engineered cell comprises a heterologous nucleotide sequence encoding a heterologous polypeptide capable of hydrolyzing sucrose into fructose and glucose which enables utilization of sucrose as sole carbon and energy source of said genetically engineered cell, and wherein the polypeptide capable of hydrolyzing sucrose into fructose and glucose is SacC Agal comprising SEQ ID NO: 13 or Bff of SEQ ID NO: 14, or a functional homologue thereof having an amino acid sequence which is at least 90% identical to any one of SEQ ID NOs: 13 or 14.
17. The method according to claim 11, wherein the genetically engineered cell is cultured in sucrose as sole carbon and energy source.
18. The method according to claim 11, wherein the genetically engineered cell comprises a heterologous nucleotide sequence encoding a major facilitator superfamily (MFS) transporter protein selected from Nec comprising SEQ ID NO: 4 and YberC comprising SEQ ID NO: 5, or a functional homologue thereof having an amino acid sequence which is at least 90% identical to any one of SEQ ID NOs: 4 or 5.
19. The method of claim 11, wherein the one or more fucosylated HMOs harvested in step (c) are further purified to a degree that the one or more HMOs can be used as a food ingredient.